MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
1800 SECURITY BASE CAMP
ACCOMMODATION 1
DESIGN OF BUILDING STRUCTURE

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
HCS	:	Hollow Core Slab
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of an Accommodation 1 Building.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of Accommodation 1 Building only.

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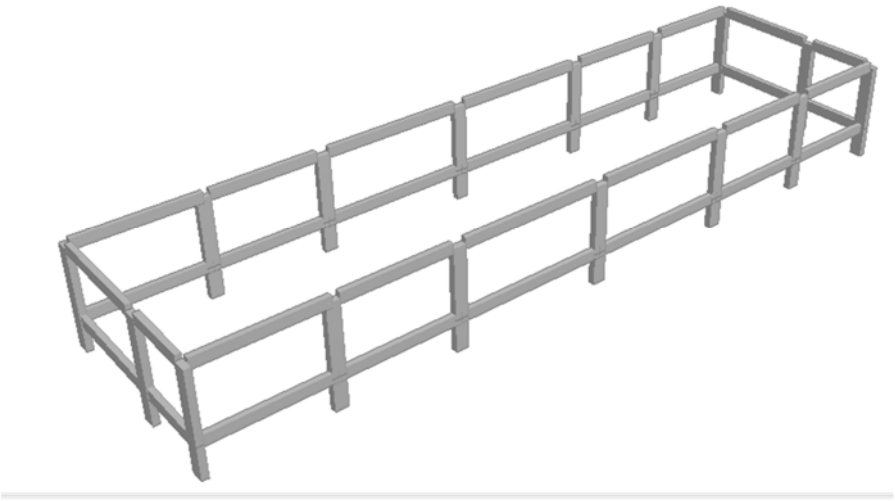
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4. DESIGN METHODOLOGY & ASSUMPTIONS

This Building Is a Single Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have Insulated Block Walls and Cast in Site Framing Beams to support the HSC (Hollow Core Slab)

The Superstructure for this Building was modelled using STAAD software. The Analysis of Frames (Cast-In-Situ Columns & Beams) has been performed meeting design requirements & keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to generate the Column Reactions at the Base that are assumed to be fixed. The Foundation design has been done by importing Column Base Reactions into Foundation Design Software.



STAAD 3D MODEL

4.1. Codes

[1] SBC-301-2007-Structural Loading & Forces.

[2] SBC-303-2007-Structural Soil & Foundations.

[3] SBC-304-2007-Structural Concrete Structures.

[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 Building Code Requirements.

[8] PCI –MNL-120-04 –Design Handbook, 6th edition.

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4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-RPT-1008.

4.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel.

5. DESIGN LOADS

5.1. Dead Load

5.1.1. Roof Dead Loads

- Provision for Electrical Lighting = 0.05 kN/m²

Ceiling

Suspended steel channel system = 0.10 kN/m²

Acoustical Fiber Tile (16mm) = 0.05 kN/m²

Sub-total = 0.2 kN/m²

- Hollow Core Slab
  (HCS-300 Topped)(To be confirmed by Vendor) = **3.88 KN/m2**

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Prestressed Hollowcore Floor

Overall Structural Depth (mm)	Self Wt- KN/M2	No. of Cores in Unit	Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m² for Finishes Characteristic Service Loads Kn/m²							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.00
			Clear Span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = **3.1 kN/m²**

Total Load =3.88+0.2+3.1= 7.2 **kN/m²**

UDL on Edge Ledger Beams (LB1) = 7.2 x 9.67/2 = **34.82 Kn/m**

Moment on all the External Beams due to Cantilever Portion of slab and Screed

= (0.15x24+(0.05+0.2)/2x20)*0.6*0.6/2 = **1 knm/m**

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5.1.2. Ground Floor Walls

- 250 tyke. Outer CMU (Partially Grouted Cell@800 c-c)

16 x 0.25 x (3.5-0.5) m clear height = **12 kN/m**

5.2. Live Load

5.2.1. Roof Live Loads

Live loads are with reference to Clause-4.2.2, MD-216-1000-EG-ST-CRT-1001

Ordinary Flat (Inaccessible) = **0.5 kN/m²**

UDL Due to Live Load on the Edge Ledger Beam (LB1& LB2) =0.5X9.67/2= **2.42 kN/m**

Moment on all the External Beams due to Live Load on

Cantilever Portion of slab = 0.5*0.6*0.6/2 = **0.09 kN/m /m**

5.3. Wind Load

Wind loads were automatically generated in STAAD as per ASCE-07. The parameters used for the wind load generation are as follows (Refer STRUCTURAL DESIGN CRITERIA NON PROCESS BUILDINGS Doc No MD-216-0000-EG-ST-CRT-0001)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

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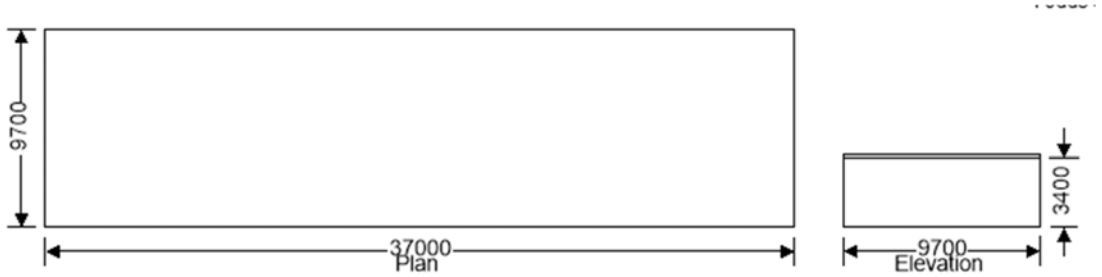
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5.4. Wind Load (Analytical Verification)

WIND LOADING (SBC-301)

Using the directional design method



Building data

Type of roof;	Flat
Length of building;	b = 37000mm
Width of building;	d = 9700 mm
Height to eaves;	H = **3400** mm
Mean height;	h = **3400** mm
Height of Parapet;	hp = 150 mm

General wind load requirements

Basic wind speed;	V = **49** m/s (or) 177 kmph (Refer Clause-4.2.6, Doc- MD-216-1000-EG-ST- CRT-1001)
Risk category;	II(As per SBC-301, Table 1.6-1)
Velocity pressure exponent coeff ;	K_d = **0.85**(As per SBC-301, Table 6.4-1)
Exposure category;	D(As per SBC-301, Table 6.4.2.3)
Enclosure classification;	Partially enclosed buildings(As per SBC-301, Section-6.2)
Internal pressure coef +ve;	GC_{pi_p} = **0.55**(As per SBC-301, Figure-7.2-1)
Internal pressure coef –ve;	GC_{pi_n} = **-0.55**(As per SBC-301, Figure-7.2-1)
Gust effect factor;	G_f = **0.85**(As per SBC-301, Section-7.2.7.1)
Importance Factor;	1(As per SBC-301, Section-9.5)

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Topography

Topography factor not significant; K_{zt} = 1.0(As per SBC-301, Section-6.4.3.2)

Velocity pressure equation; $q = 0.0473x10^{-3}Kn/m^2 \times K_z \times K_{zt} \times K_d \times V^2$ xI

V in Km/hr(As per SBC-301, Section-7.2.8)

Velocity pressures table

z (mm)	K_z (SBC-301,Table 7.2-2)	q_z (kN/m^2)
3400	1.03	1.29
3550	1.03	1.29

Pressures and forces

Net pressure; $p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$;

Net force; $F_w = p \times A_{ref}$;

Walls load case 1 - Wind 0, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.40	0.80	1.29	0.17	125.75	21.06
B	3.40	-0.50	1.29	-1.26	125.80	-158.05
C	3.40	-0.70	1.29	-1.48	32.98	-48.66
D	3.40	-0.70	1.29	-1.48	32.98	-48.66

Walls load case 2 - Wind 0, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.40	0.80	1.29	1.58	125.75	199.93
B	3.40	-0.50	1.29	0.16	125.80	20.26
C	3.40	-0.70	1.29	-0.06	32.98	-1.91
D	3.40	-0.70	1.29	-0.06	32.98	-1.91

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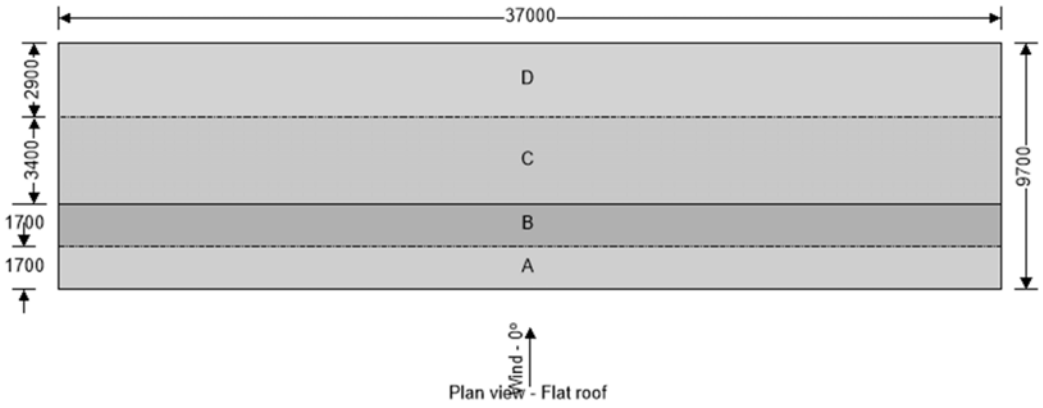
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Walls load case 3 - Wind 90, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m²)	Net pressure p (kN/m²)	Area A_{ref} (m²)	Net force F_w (kN)
A	3.40	0.80	1.29	0.17	32.97	5.52
B	3.40	-0.24	1.29	-0.94	32.97	-30.93
C	3.40	-0.70	1.29	-1.48	125.80	-185.61
D	3.40	-0.70	1.29	-1.48	125.80	-185.61

Walls load case 4 - Wind 90, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m²)	Net pressure p (kN/m²)	Area A_{ref} (m²)	Net force F_w (kN)
A	3.40	0.80	1.29	1.58	32.97	52.25
B	3.40	-0.24	1.29	0.48	32.97	15.81
C	3.40	-0.70	1.29	-0.06	125.80	-7.29
D	3.40	-0.70	1.29	-0.06	125.80	-7.29

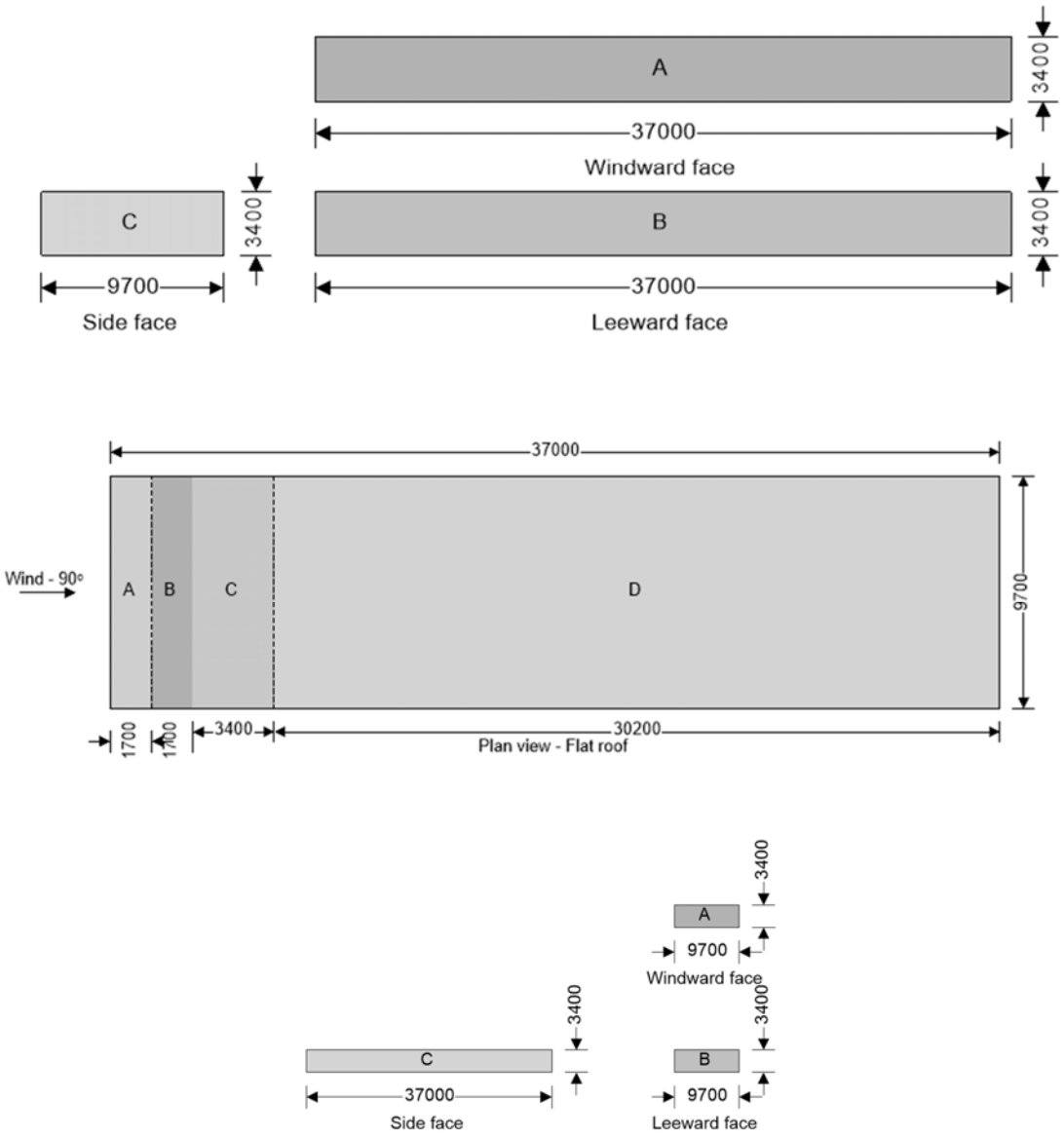


Plan view - Flat roof

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So from the above analysis, the Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

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5.5. Seismic Load

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

Where S_{D_1} Damped Design Spectral Response Acceleration

Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

For Ordinary reinforced concrete moment frame,

Response Modification Factor, Rx = 2.5 (As per SBC-301, Table 10.9.3.2)

Response Modification Factor, Rz = 2.5 (As per SBC-301, Table 10.2)

System Over Strength Factor, Omega = 3 (As per SBC-301, Table 10.2)

Deflection Amplification factor, Cd = 2.5 (As per SBC-301, Table 10.2)

5.6. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

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6. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load

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7. CRACKED SECTIONS

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

For Beams:	I = 0.35Ig
For Columns:	I = 0.70Ig
For Walls:	
Un-cracked	I = 0.70Ig
Cracked	I = 0.35Ig
For Slabs	I = 0.25Ig

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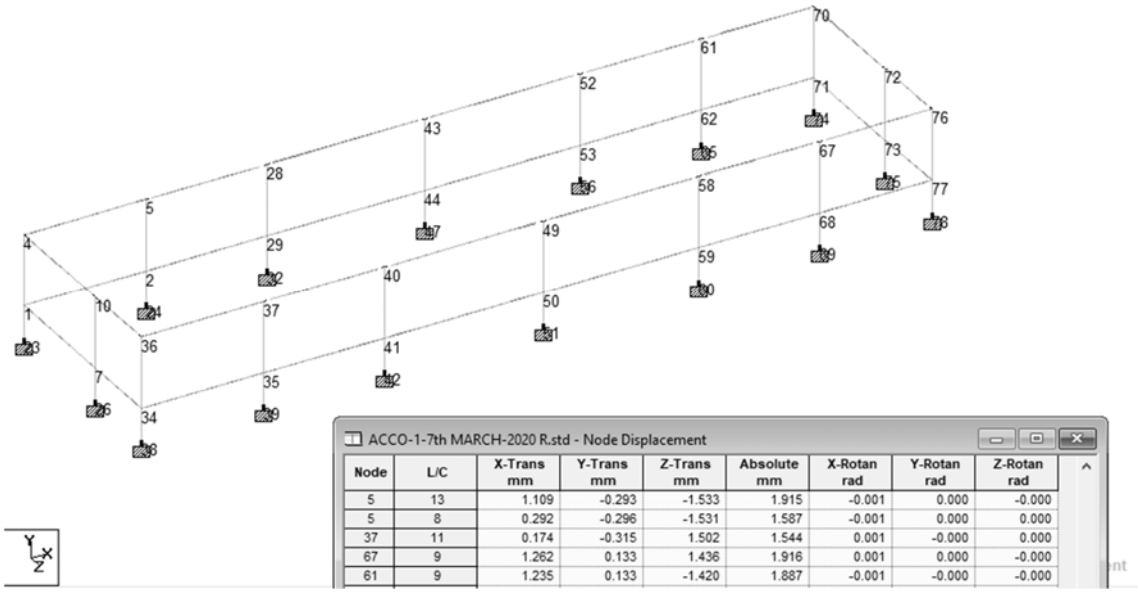
8. DEFLECTIONS/DRIFT

8.1. Maximum Allowable Deflection

L/240: for member load with live load plus dead load

L/600: for members supporting masonry under total loads

Where, L = span of member between support



Maximum attained Drift as per Analysis from above figure = 25mm

Allowable Drift as per Table-10.12, SBC-301-2007 is H/100 = 3300/100 = 33mm

Hence the Structure is safe considering drift check…

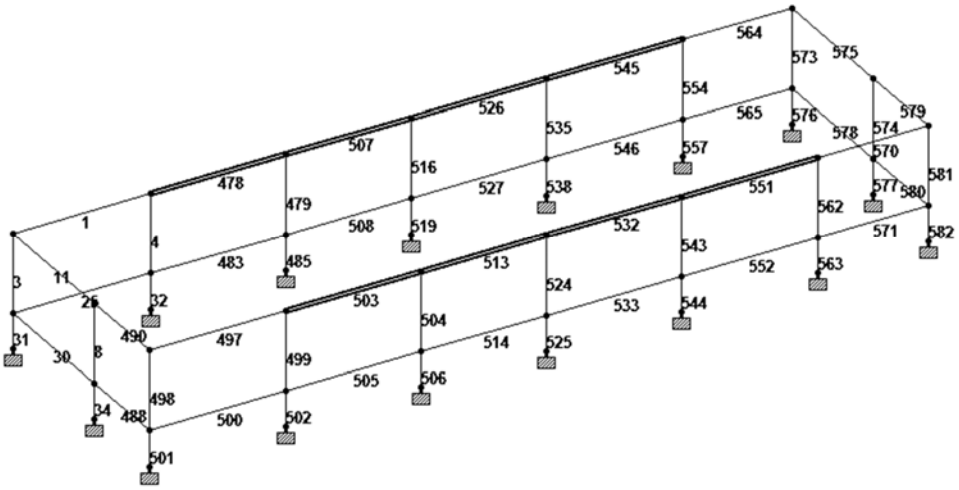
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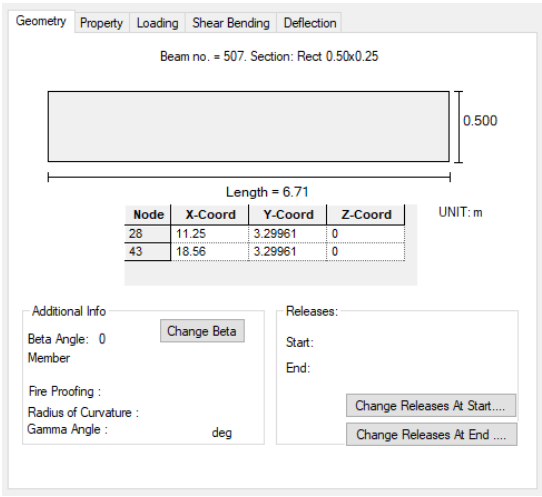
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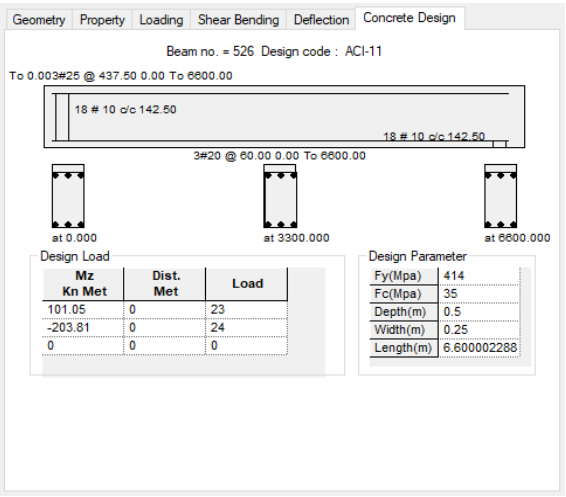
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9. CONCRETE ROOF BEAM DESIGN

9.1. Design of Ledger Beam(LB1)







Design of Ledger Beam (LB1)

Maximum Moment required from Analysis = 203 **Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

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Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ =0.9

Ast1.req = **1350 mm²**

Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **406mm²**

Maximum Moment required from Analysis = **101 Knm (Sagging)**

Ast2.req = **426 mm²**

Hence Provide 3-25 dia-Full Straight Bars Top & 3-25 dia-Full Straight Bottom considering long term deflection as shown in section 8.1.1

Ast.pro**>>>>>>(Ast.min, Ast.req)**

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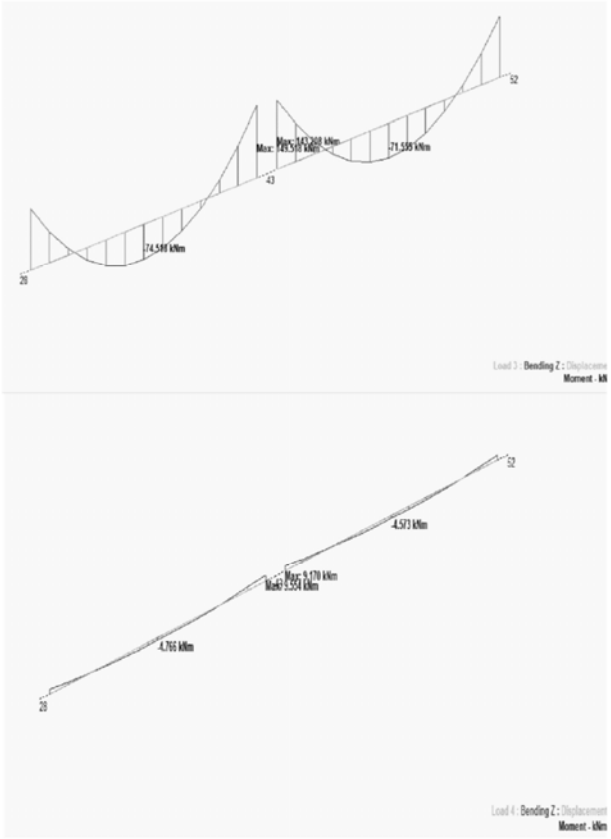
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9.1.1. Long Term Deflection Check for LB1

INPUT DATA :

Conc. Compressive stress, fc`	=	35.00	MPa
Reinforcement bar tensile stress, fy	=	414.00	MPa
Modulus Elasticity of concrete,	=	27,806	Mpa
Modulus Elasticity of steel, *Es*	=	200,000	Mpa
Unit wt. of concrete, Wconc	=	24.00	kN/m^3
Unit wt. of steel	=	77.00	kN/m^3
Depth of beam, h	=	500	mm
width of beam, b	=	250	mm
Length of beam, L	=	7.00	m
Bottom bar diam., db	=	25.00	mm
area of 1 bar Ab	=	490.87	mm^2
no. of bottom bars	=	3.00	pcs.
Top bar diam., db	=	25.00	mm
area of 1 bar Ab	=	490.87	mm^2
no. of top bars	=	3.00	pcs.
Stirrups diam. Dv	=	10.00	mm
area of 1 bar Av	=	78.54	mm^2
concrete cover, cc	=	40.00	mm
$d_{eff\ bot}$ (@ bottom), d	=	437.50	mm
Mdead	=	77.00	kN-m
M live	=	5.00	kN-m



CHECK DEFLECTION: $M_{d+\ell}$ = 82.00 kN-m

$M_{sus} = M_d + 0.50M_\ell$ = 79.50 kN-m

(FROM STAAD RESULTS)

modulus of rupture, modulus of elasticity, modular ratio

$fr = 0.62\lambda\sqrt{fc'}$ = 3.67 Mpa *?ACI(318M-08) Sec. 9.5.2.3*

$n_s = \frac{Es}{Ec}$ = 7.19

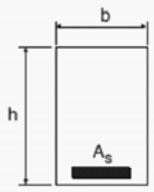
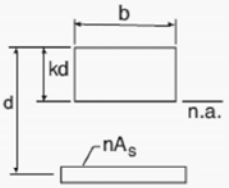
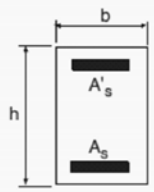
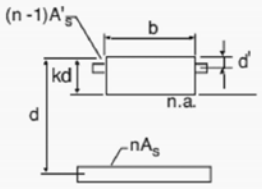
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gross and crack section of moment of inertia

Table 10-2 Gross and Cracked Moment of Inertia of Rectangular and Flanged Section

Gross Section	Cracked Transformed Section	Gross and Cracked Moment of Inertia
	 Without compression steel	$n = \frac{E_s}{E_c}$ $B = \frac{b}{(nA_s)}$ $I_g = \frac{bh^3}{12}$ Without compression steel $kd = \left(\sqrt{2dB+1}-1\right)/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2$
	 With compression steel	With compression steel $r = (n-1)A'_s/(nA_s)$ $kd = \left[\sqrt{2dB(1+rd'/d)+(1+r)^2}-(1+r)\right]/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2 + (n-1)A'_s(kd-d')^2$

$$I_g = \frac{bh^3}{12} = 2.60E+09 \text{ mm?}$$

$$B = \frac{b}{nAs} = 0.02 \text{ mm/mm²}$$

$$kd = \left(\sqrt{2dB+1}-1\right)/B = 154.78 \text{ mm}$$

$$I_{cr} = \frac{b(kd)^3}{3} + nAs(d-kd)^2 = 1.16E+09 \text{ mm?}$$

$$\frac{I_g}{I_{cr}} = 2.25$$

effective moment of inertia

$$M_{cr} = \frac{frI_g}{y_t} = 38.21 \text{ kN-m}$$ *?ACI(318M-08) Sec. 9.5.2.3*

under dead load

$$\frac{M_{cr}}{M_d} = 0.50$$

$$(I_e)_d = \left(\frac{M_{cr}}{M_d}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_d}\right)^3\right] I_{cr} = 1.33E+09 \text{ mm?}$$ *?ACI(318M-08) Sec. 9.5.2.3*

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under sustained load

$$\frac{M_{cr}}{M_{sus}} = 0.48$$

$$(I_e)_{sus} = \left(\frac{M_{cr}}{M_{sus}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{sus}}\right)^3\right] I_{cr} = 1.32E+09 \text{ mm?}$$?ACI(318M-08) Sec. 9.5.2.3

under dead + live load

$$\frac{M_{cr}}{M_{d+\ell}} = 0.47$$

$$(I_e)_{d+\ell} = \left(\frac{M_{cr}}{M_{d+\ell}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{d+\ell}}\right)^3\right] I_{cr} = 1.30E+09 \text{ mm?}$$?ACI(318M-08) Sec. 9.5.2.3

initial or short-term deflection

$K = 1.00$

$$(\Delta_i)_d = \frac{K(5/48)M_d\ell^2}{Ec(I_e)_d} = 10.607 \text{ mm}$$

$$(\Delta_i)_{sus} = \frac{K(5/48)M_{sus}\ell^2}{Ec(I_e)_{sus}} = 11.086 \text{ mm}$$

$$(\Delta_i)_{d+\ell} = \frac{K(5/48)M_{d+\ell}\ell^2}{Ec(I_e)_{d+\ell}} = 11.559 \text{ mm}$$

$$(\Delta_i)_\ell = (\Delta_i)_{d+\ell} - (\Delta_i)_d = 0.953 \text{ mm}$$

Table 10-3 Deflection Coeffcient K

		K
1.	Cantilevers (deflection due to rotation at supports not included)	2.40
2.	Simple beams	1.0
3.	Continuous beams	1.2-0.2 M_o/M_a
4.	Fixed-hinged beams (midspan deflection)	0.80
5.	Fixed-hinged beams (maximum deflection using maximum moment)	0.74
6.	Fixed-fixed beams	0.60

For other types of loading, K values are given in Ref. 10.2.

M_o = Simple span moment at midspan $\left(\frac{w\ell^2}{8}\right)$

M_a = Net midspan moment.

TABLE 9.5(b) — MAXIMUM PERMISSIBLE COMPUTED DEFLECTIONS

Type of member	Deflection to be considered	Deflection limitation
Flat roofs not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/180^*$
Floors not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/360$
Roof or floor construction supporting or attached to nonstructural elements likely to be damaged by large deflections	That part of the total deflection occurring after attachment of nonstructural elements (sum of the long-term deflection due to all sustained loads and the immediate deflection due to any additional live load)†	$\ell/480^‡$
Roof or floor construction supporting or attached to nonstructural elements not likely to be damaged by large deflections		$\ell/240^§$

*Limit not intended to safeguard against ponding. Ponding should be checked by suitable calculations of deflection, including added deflections due to ponded water, and considering long-term effects of all sustained loads, camber, construction tolerances, and reliability of provisions for drainage.

†Long-term deflection shall be determined in accordance with 9.5.2.5 or 9.5.4.3, but may be reduced by amount of deflection calculated to occur before attachment of nonstructural elements. This amount shall be determined on basis of accepted engineering data relating to time-deflection characteristics of members similar to those being considered.

‡Limit may be exceeded if adequate measures are taken to prevent damage to supported or attached elements.

§Limit shall not be greater than tolerance provided for nonstructural elements. Limit may be exceeded if camber is provided so that total deflection minus camber does not exceed limit.

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$(\Delta_i)_\ell \le \frac{\ell}{180}$ = 38.89 mm > 0.95 mm *ok!*

$(\Delta_i)_\ell \le \frac{\ell}{360}$ = 19.44 mm > 0.95 mm *ok!*

additional long-term deflection

ξ = 2.00

5 years or more	2.0
12 months	1.4
6 months	1.2
3 months	1.0

$\rho = \frac{As}{bd}$ = 0.0135

$\lambda_\Delta = \frac{\xi}{1+50\rho'}$ = 1.20 *?ACI(318M-08) Sec. 9.5.2.5*

$\Delta_{cp} + \Delta_{sh} = \lambda(\Delta_i)_{sus}$ = 13.25 mm

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell$ = 14.20 mm

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell \le \frac{\ell}{480}$ = 14.58 mm > 14.20 mm *ok!*

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell \le \frac{\ell}{240}$ = 29.17 mm > 14.20 mm *ok!*

9.1.2. Staad Output for LB1

ACI 318-11 BEAM NO. 507 DESIGN RESULTS

LEN - 6710. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	60.	3 - 20MM	0.	6710.	YES	YES
2	438.	3 - 25MM	0.	6710.	YES	YES

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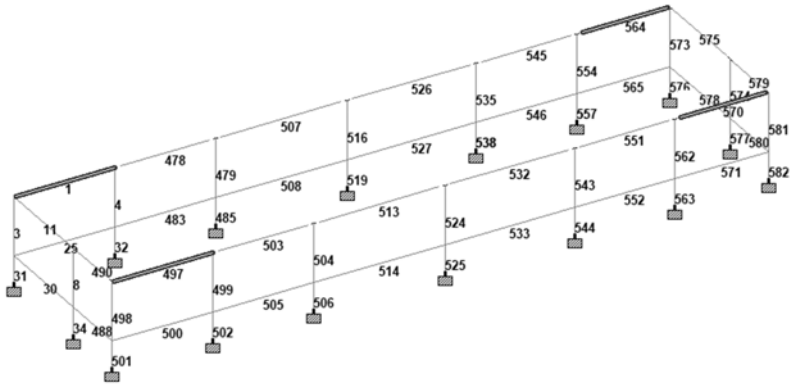
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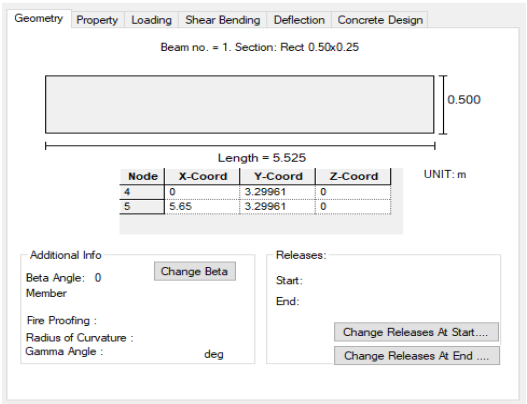
```
B E A M  N O.   507 D E S I G N  R E S U L T S - SHEAR

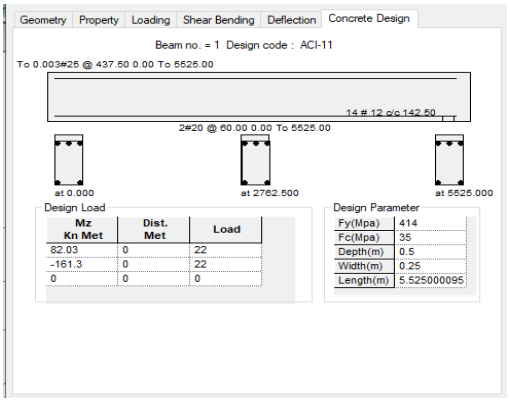

AT START SUPPORT - Vu=  114.37 KNS  Vc=  123.50 KNS  Vs=   29.00 KNS
Tu=    4.60 KN-MET  Tc=    4.2 KN-MET  Ts=    6.1 KN-MET  LOAD    25
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2415. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.52 SQ.CM.

AT END   SUPPORT - Vu=  116.96 KNS  Vc=  114.48 KNS  Vs=   41.47 KNS
Tu=    4.68 KN-MET  Tc=    4.0 KN-MET  Ts=    6.2 KN-MET  LOAD    28
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2415. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.54 SQ.CM.
```

9.2. Design of Ledger Beam(LB2)







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Design of Ledger Beam (LB2)

Maximum Moment required from Analysis = 161 **Knm (Hogging)**

Therefore Ast.req = $$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ =0.9

Ast1.req = **1100 mm²**

Ast.min= (As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f_c'}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **406mm²**

Maximum Moment required from Analysis = **82 Knm (Sagging)**

Hence Provide 3-25 dia-Full Straight Bars Top & 3-20 dia-Full Straight Bottom

Ast.pro **>>>>>>(Ast.min, Ast.req)**

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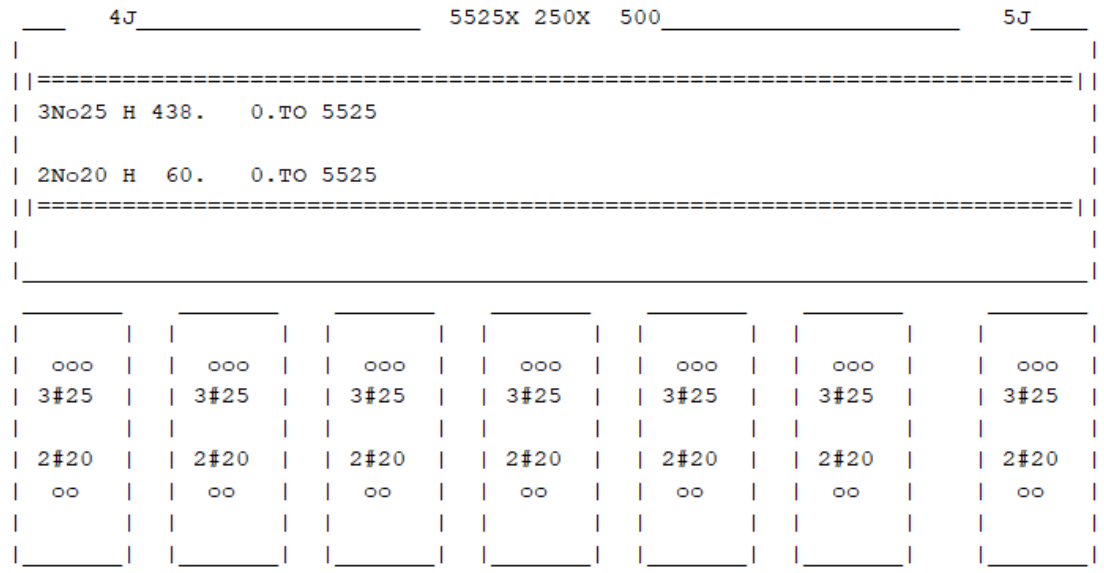
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9.2.1. Staad Output For LB2

```
        B E A M  N O.     1 D E S I G N  R E S U L T S - SHEAR


  AT START SUPPORT - Vu=   67.71 KNS  Tu=   29.3 KN-MET
                     Vc=  116.1 KNS, ACI 318:CLAUSE 11.6.3.1
LOAD    27 TORSION VALUE TOO HIGH, INCREASE MEMBER SIZE.
  AT END   SUPPORT - Vu=  102.70 KNS  Tu=   24.8 KN-MET
                     Vc=  120.1 KNS, ACI 318:CLAUSE 11.6.3.1
LOAD    27 TORSION VALUE TOO HIGH, INCREASE MEMBER SIZE.
```

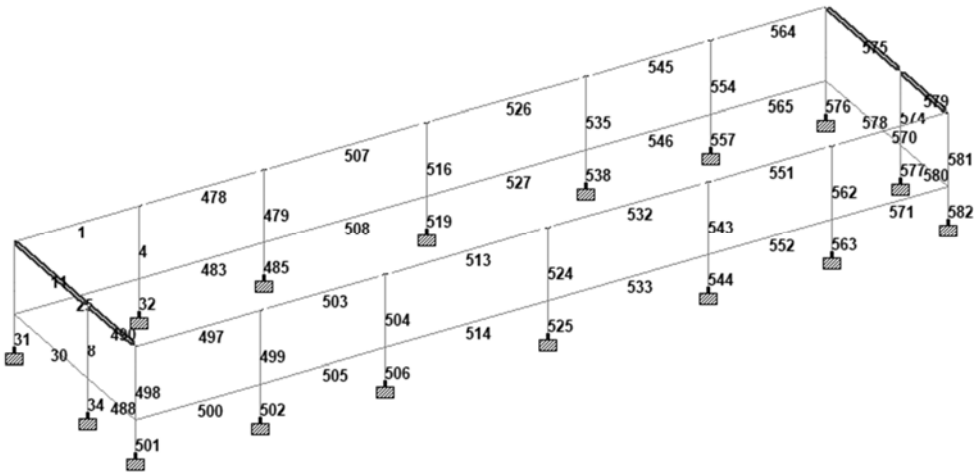


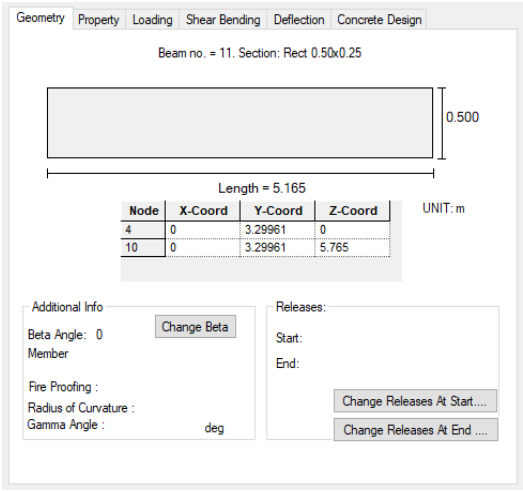
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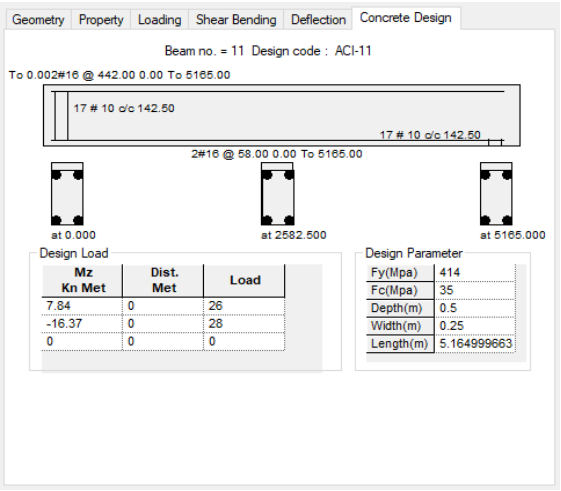
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9.3. Design of Beam-(B1)







Ast.min= **406mm²**

Hence Provide 3-16 dia-Full Straight Bars Top & Bottom

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9.3.1. Staad Output For B1

```
        ACI 318-11   BEAM  NO.    11   DESIGN RESULTS
        ==============================================

LEN -  5165. MM  FY -  414.  FC -  35.  MPA, SIZE -  250. X  500. MMS

LEVEL     HEIGHT      BAR INFO        FROM          TO          ANCHOR
            (MM)                        (MM)          (MM)        STA  END
_____________________________________________________________________________


  1          58.      2 - 16MM          0.          5165.         YES  YES
  2         442.      2 - 16MM          0.          5165.         YES  YES


       B E A M  N O.    11 D E S I G N  R E S U L T S - SHEAR


 AT START SUPPORT - Vu=     3.35 KNS  Vc=  103.89 KNS  Vs=     0.00 KNS
 Tu=    14.94 KN-MET  Tc=     4.0 KN-MET  Ts=   19.9 KN-MET  LOAD     26
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2143. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    4.92 SQ.CM.

 AT END   SUPPORT - Vu=    11.83 KNS  Vc=  104.11 KNS  Vs=     0.00 KNS
 Tu=     9.58 KN-MET  Tc=     4.0 KN-MET  Ts=   12.8 KN-MET  LOAD     26
 STIRRUPS ARE REQUIRED FOR TORSION.
```


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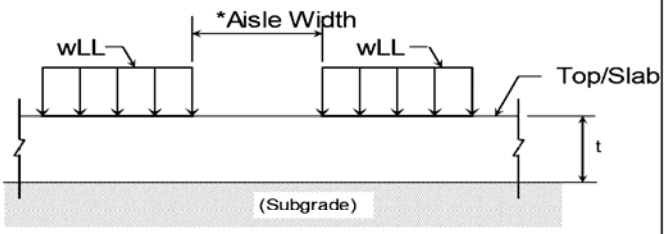
10. CONCRETE GRADE SLAB DESIGN

CONCRETE SLAB ON GRADE ANALYSIS

For Slab Subjected to Stationary Uniformly Distributed Live Loads

Input Data:

Slab Thickness, t =	3.937	in.	100.000	mm		
Concrete Strength, f 'c =	5000	psi	35.000	kpa		
Subgrade Modulus, k =	39.7866	pci	10800	Kn/m3	(AS PER SECTION-7.7-SOIL REPORT)	
Factor of Safety, FS =	1.500		1.500			
Uniform Live Load, wLL =	104.45	psf	5.00	kpa		



Concrete Slab on Grade with Uniform Loads

Results:

Design Parameters:

Modulus of Rupture, MR =	636.40	psi	4387.79	kpa	MR = 9*SQRT(f 'c)
ow. Bending Stress, Fb =	424.26	psi	2925.19	kpa	Fb = MR/FS
Modulus of Elasticity, Ec =	4030509	psi	27789329	kpa	Ec = 57000*SQRT(f 'c)
Poisson's Ratio, μ =	0.15		0.15		μ = 0.15 (assumed for concrete)
Radius of Stiffness, Lr =	26.94	in.	684.37	mm	Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25
Critical Aisle Width, Wcr =	4.96	ft.	1511.77	mm	Wcr = (2.209*Lr)/12

Stationary Uniformly Distributed Live Loads:

wLL(allow) =	682.05	psf	wLL(allow) = 257.876*Fb*SQRT(k*t/Ec)
	32.65	kpa	**wLL(allow) >= wLL, O.K.**

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 140 mm2/M

Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area

Reinforcement type	f_y, psi	Minimum reinforcement ratio	
Deformed bars	< 60,000	0.0020	
Deformed bars or welded wire reinforcement	≥ 60,000	Greater of:	$\frac{0.0018 \times 60,000}{f_y}$
			0.0014

Provide WWF-8mm Dia@200 c-c Both Ways-TOP LAYER

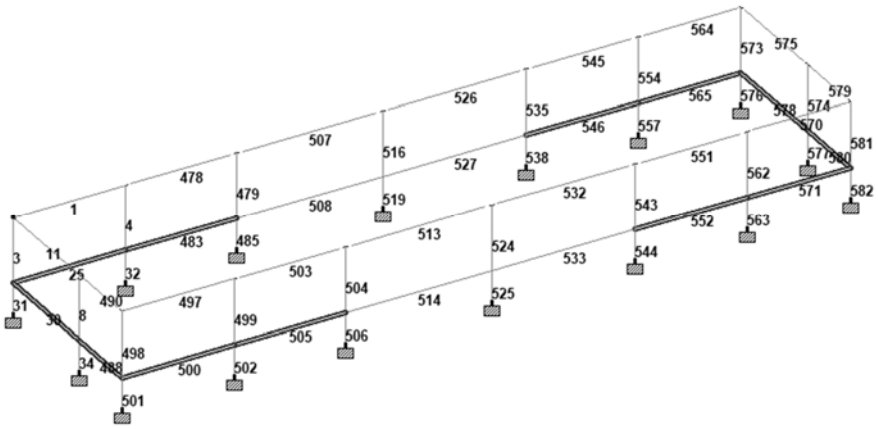
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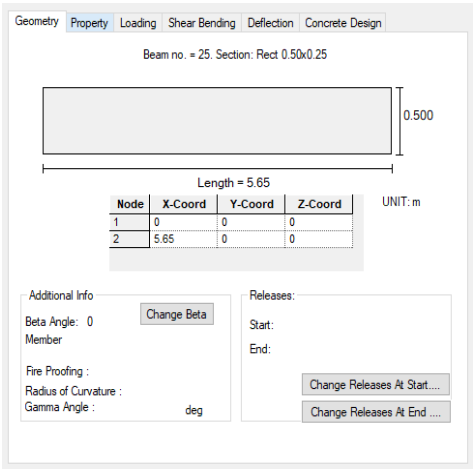
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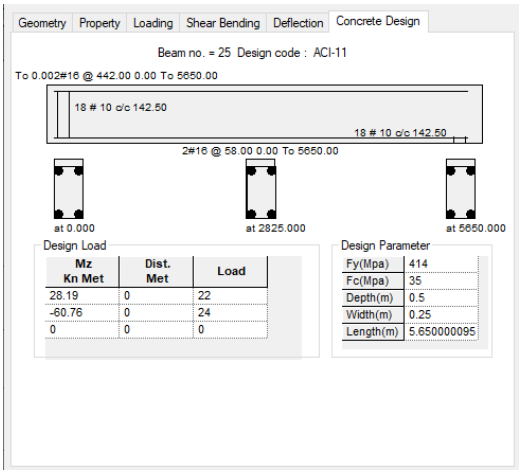
11. DESIGN OF GRADE BEAM

11.1. Grade Beam(GB1)



MOMENT ENVELOPE FOR BEAMS GB1





However Provide 3-20 dia-Full Straight Bars Top & 3-16 dia-Full Straight Bars Bottom considering minimum reinforcement requirements.

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11.1.1. Staad Output for GB1

```
        ACI 318-11   BEAM  NO.    25   DESIGN RESULTS
        ==============================================

LEN -  5650. MM  FY -  414.  FC -  35.  MPA, SIZE -  250. X  500. MMS

LEVEL     HEIGHT      BAR INFO        FROM          TO          ANCHOR
            (MM)                        (MM)         (MM)       STA  END
_____________________________________________________________________

  1         58.     2 - 16MM          0.          5650.        YES  YES
  2        442.     2 - 16MM          0.          5650.        YES  YES


        B E A M  N O.    25 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=   40.94 KNS  Vc=  107.75 KNS  Vs=    0.00 KNS
Tu=   13.36 KN-MET  Tc=    3.8 KN-MET  Ts=   17.8 KN-MET  LOAD    27
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2385. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   4.40 SQ.CM.

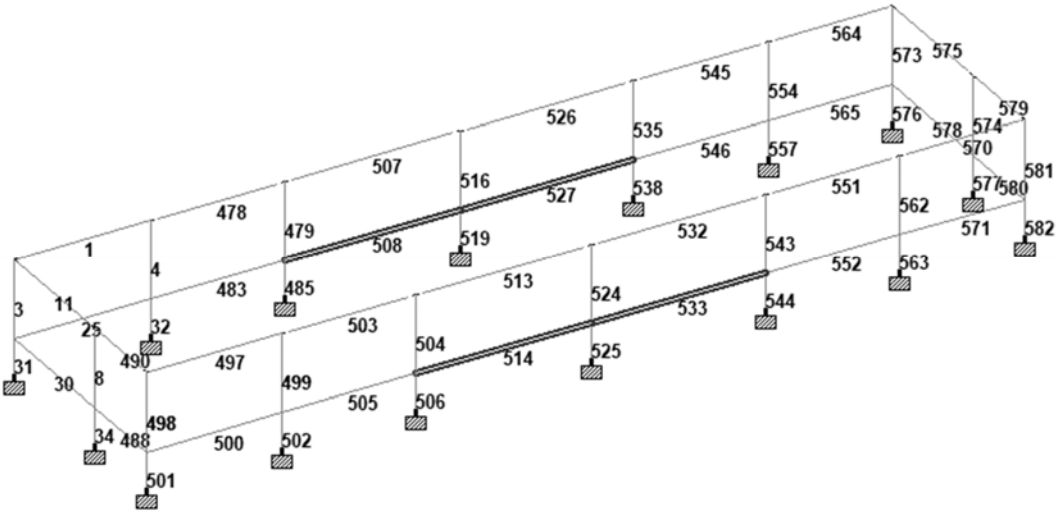
AT END   SUPPORT - Vu=   44.65 KNS  Vc=  107.75 KNS  Vs=    0.00 KNS
Tu=   13.36 KN-MET  Tc=    3.8 KN-MET  Ts=   17.8 KN-MET  LOAD    27
STIRRUPS ARE REQUIRED FOR TORSION.
```


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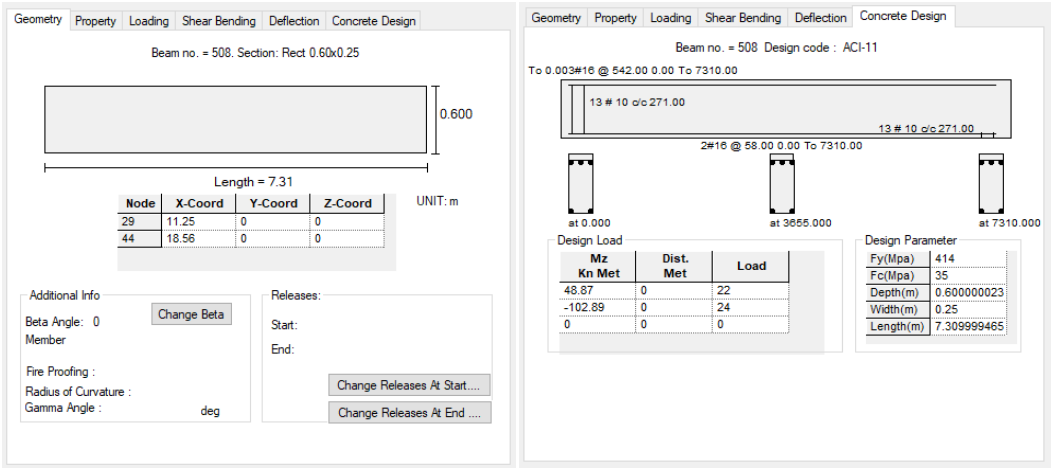
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11.2. Grade Beam(GB2)



MOMENT ENVELOPE FOR BEAMS GB2



Maximum Moment required from Analysis = 102 **Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-50-10-16/2=432mm, Fc'=35 Mpa & ϕ=0.9

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Ast.min= $A_{s,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d$ (10-3)

and not less than $1.4 b_w d / f_y$.

Ast.min= **481mm²**

Maximum Moment required from Analysis = **48 Knm (Sagging)**

Hence Provide 3-20 dia-Full Straight Bars Top & 3-20 dia-Full Straight Bars Bottom

Ast.pro **>>>>>>(Ast.min, Ast.req)**

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11.2.1. Staad Output for GB2

```
          ACI 318-11   BEAM  NO.   508   DESIGN RESULTS
          ==============================================

LEN -  7310. MM  FY -  414.  FC -  35.  MPA, SIZE -  250. X  600. MMS

LEVEL     HEIGHT      BAR INFO        FROM          TO          ANCHOR
            (MM)                        (MM)          (MM)      STA  END
_____________________________________________________________________

  1         58.      2 - 16MM          0.          7310.        YES  YES
  2        542.      3 - 16MM          0.          7310.        YES  YES


        B E A M  N O.   508 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=   49.14 KNS  Vc=  123.71 KNS  Vs=    0.00 KNS
Tu=    0.02 KN-MET  Tc=    4.6 KN-MET  Ts=    0.0 KN-MET  LOAD     9
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 271. MM  C/C FOR 3115. MM

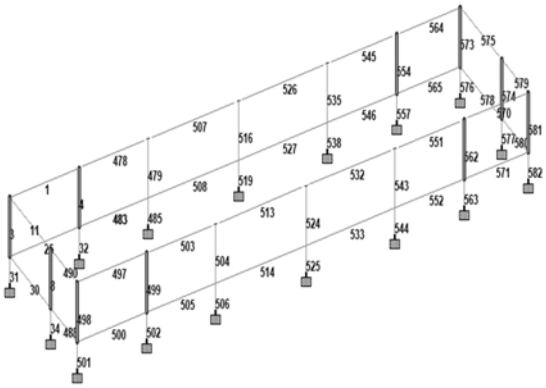
AT END   SUPPORT - Vu=   47.69 KNS  Vc=  123.71 KNS  Vs=    0.00 KNS
Tu=    0.02 KN-MET  Tc=    4.6 KN-MET  Ts=    0.0 KN-MET  LOAD     9
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 271. MM  C/C FOR 3115. MM
```


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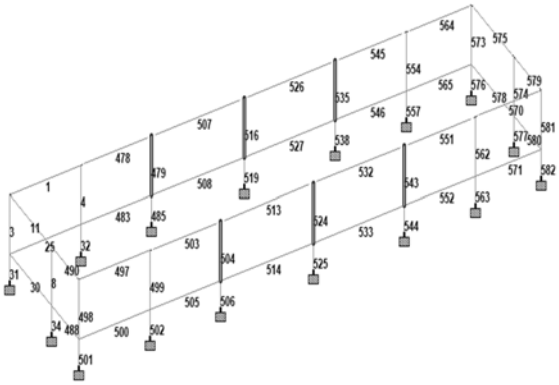
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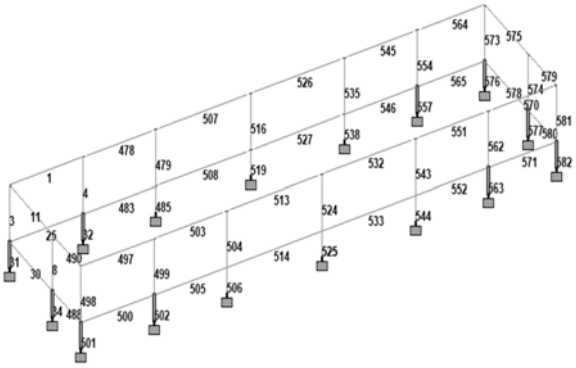
12. CONCRETE COLUMN DESIGN



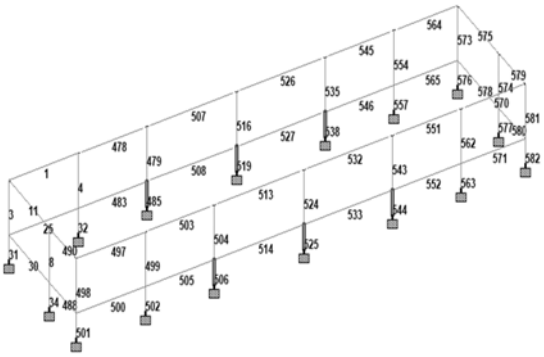
Column numbers C1



Column numbers C2



Column numbers CR

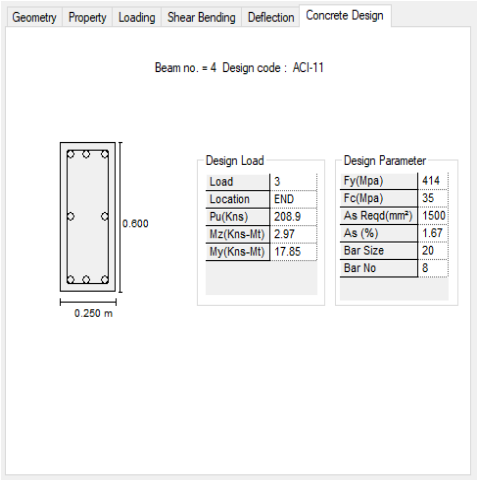


Column numbers CR1

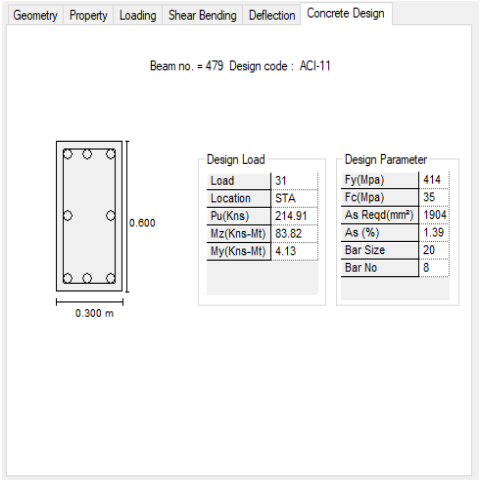
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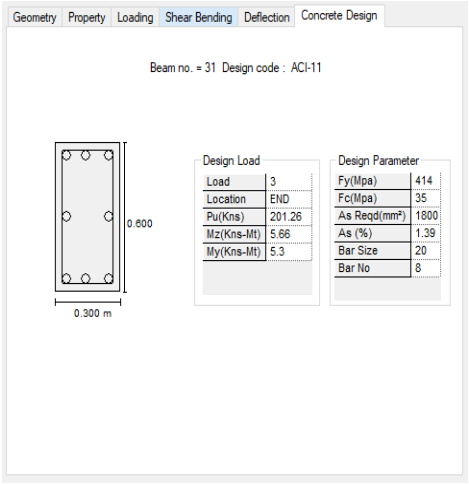
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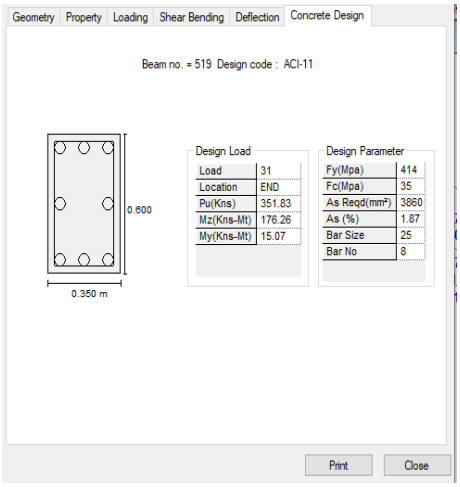
Column Design-C1



Column Design-C2



Column Design-CR (Column Neck)



Column Design- CR1 (Column Neck)

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12.1. Staad Output For Column C1, C2, CR & CR1

```
      ACI 318-11   COLUMN  NO.     4   DESIGN RESULTS
      ================================================


FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 600.0 X 250.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  1500.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-------------------------------------------------------------

  8 - 20 MM              1.676        3      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 250.00 MM



      ACI 318-11   COLUMN  NO.   479   DESIGN RESULTS
      ================================================


FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 600.0 X 300.0 MMS, TIED
        AREA OF STEEL REQUIRED =  2185.2  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-------------------------------------------------------------

  8 - 20 MM              1.396        31      STA     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 300.00 MM
```


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```
        ACI 318-11   COLUMN  NO.    31   DESIGN RESULTS
        ================================================


  FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 600.0 X 300.0 MMS, TIED
          ONLY MINIMUM STEEL IS REQUIRED.
          AREA OF STEEL REQUIRED =  1800.0  SQ. MM

  BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
  ------------------------------------------------------------

    4 - 25 MM               1.091        3      END     0.650
  (PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
  TIE BAR NUMBER   10 SPACING 300.00 MM


     ACI 318-11   COLUMN  NO.   519   DESIGN RESULTS
     ================================================


FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 600.0 X 350.0 MMS, TIED
        AREA OF STEEL REQUIRED =  3859.8  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
------------------------------------------------------------

  8 - 25 MM               1.870       31      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 320.00 MM
```


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13. CANTILEVER SLAB DESIGN

DESIGN OF CANTILEVER SLAB

Geometrical Properties			Material Properties			
Span to center of support "sp_1"	0.8	met	Weight of reinforced concrete "γ_r	24		kN/m³
Clear Span	0.6	met	Strength of concrete "f'_c"	35		N/mm²
Clear span + depth of slab "sp_2"	0.75	met	Strength of steel "f_y"	420		N/mm²
Max. eff span le = min (sp_1;sp_2)	0.8	met	Clear cover to concrete("c")	20		mm
Width of Slab "b"	1000	mm	Flexure strength reduction factor	0.9		
Thickness of slab "t"	150	mm	Shear strength reduction factor	0.75		
Eff. Depth "d"	125		Modulus of Elasticity of steel "Es"	2E+05		N/mm²
Loading			Modulus of Elasticity of concrete	27806		N/mm²
Self Weight of slab	3.6	kN/m²	Modular Ratio "n = Es/Ec"	7.19		
Floor Finish	3.5	kN/m²	**Provided Reinforcement**			
Wall load	0	kN/m²	Top layer Nos.	6	Φ 10	mm
Total dead Load "U_{dl}"	7.1	kN/m²	Bottom layer Nos.	0	Φ 12	mm
Live load	1	kN/m²	Area of Top steel "$A_{s\ t}$"		468	mm²
Total Ultimate Load "$Uu_{,tot}$"(1.2D+1.6L)	10.12	kN/m²	Area of Bottom steel "$A_{s\ b}$"		0	mm²
Moment due to Overhang Point Load(0.125*0.15*24*0.6)	0.27	kNm	"db" = 10 mm			
Calculations						
Shear force Vu = U*le	**8.096**	**kN**				
Neg. Moment Mu = U*le²/2	**3.51**	**kNm**				

Check for Section capacity

Depth of comp. block "a" = $A_{s\ t}*f_y/(0.85*f'_c*b)$	6.61	mm						
Section capacity ΦMn = $\Phi*A_{s\ t}*f_y*(d-a/2)$	**21.94**	**kNm**	**O.K**	**>**	**3.51**	**kNm**	(Mu)	

Shear requirements

Shear capacity ΦVc = Φ*0.17*√f'c*b*d	**94.29**	**kN**	**O.K**	**>**	**8.096**	**kN**	(Vu)	

Serviceability requirements

d' = $c+d_b/2$	26	mm
B = $b/(n*A_{s\ t})$	0.297	
r = $(n-1)*A_{s\ b}/(n*A_{s\ t})$	0	
k_d = $(\sqrt{(2*d*B(1+r*d'/d)+(1+r)^2)}-(1+r))/B$	29.01	mm
I_{cr} = $bkd^3/3+n*A_{s\ t}*(d-k_d)^2+(n-1)*A_{s\ b}*(k_d-d'$	3.915E+07	mm⁴
Calculate Gross moment of inertia		
Area of slab A = t*b	2E+05	mm²
Depth of neutral axis "y"	75	mm
Distance of top fiber from N.A "y_t"	75	mm
Gross moment of inertia "I_g"	2.813E+08	mm⁴
Allowable stress fr = 0.62√f'c	3.67	mm²
Cracking moment M_{cr} = $fr*I_g/y_t$	13.75	kNm

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Service moments

Moment due to dead load "M_{dl}"	2.54	kNm
Moment due to live load "Mll"	0.32	kNm
Total Moment Ms = $M_{dl}+M_{ll}$	2.86	kNm

Deflection due to dead + live loads

R(dl+ll) = M_{cr}/Ms	1.000		
$I_e(dl+ll) = R^3*I_g+(1-R^3)*I_{cr}$	2.813E+08	mm^4	
$\delta = K*5/48*Ms*le^2/(Ec*I_e)$	0.08	mm	**K = 3.2 (conservative)**

Deflection due to dead load

R(dl) = M_{cr}/Md	5.411	
$I_e(dl) = R^3*I_g+(1-R^3)*I_{cr}$	3.839E+10	mm^4
$\delta = K*5/48*Md*le^2/(Ec*I_e)$	0.00	mm

Deflection due to live load

Live load deflection = dl+ll deflection - dl deflection 0.08 mm

Allowable immediate deflection due to Live load =
per ACI 318 Tab 9.5b **$I_e/360$ 2.22 mm > 0.08 mm O.K**

Long term deflection

% of liveload sustained "SL"	50	%
$M_{sus} = M_{dl}+SL*M_{ll}$	2.70	kNm
R(sus) = $M_{cr}/Msus$	5.091	
$I_e(sus) = R^3*I_g+(1-R^3)*I_{cr}$	3.198E+10	
$\delta = K*5/48*Msus*le^2/(Ec*I_e)$	0.00	mm

Magnification factor for creep and shrinkage

$\lambda_\Delta = \xi / (1+50*\rho')$ 2.000

Where:-

ξ = 2 for 5 years or more cl.9.5.2.5 ACI 318

$\rho' = A_{s\ b}/(b*d)$ 0

Total deflection = λ_Δ*sus deflection + ll deflection 0.079 mm

Allowable deflection = $I_e/240$ 3.333 mm > 0.079 mm O.K

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14. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 350 Kpa. (Refer Doc-MD-216-3000-EG-CV-RPT-1008 for Details)

TABLE 10: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^2)
Village Area	BH-21	Village	Isolated Square	1.20	1.0 x 1.0	1000	350	350	25	42,000	32,200
				1.50		1100	500	500	25	60,000	38,900
				1.20	2.0 x 2.0	700	580	580	25	69,600	37,500
				1.50		850	750	750	25	90,000	44,200
				1.20	3.0 x 3.0	600	750	600	25	72,000	42,800
				1.50		700	930	700	25	84,000	49,500
				1.20	4.0 x 4.0	550	910	550	25	66,000	48,100
				1.50		625	1090	625	25	75,000	54,800
				1.20	5.0 x 5.0	475	1060	475	25	57,000	53,400
				1.50		550	1240	550	25	66,000	60,200

Refer Attachment- II for Foundation Design calculations

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REFERENCES

1.) BUILDING PLANS	MD-216-1800-EG-AR-GAD-3152
2.) OVERALL LAYOUT PLAN	MD-216-1800-EG-GAD-1565
3.) LAYOUT & DETAILS OF FOUNDATION, COLUMN & GRADE BEAM	MD-216-1800-EG-CV-GAD-1542
4.) LAYOUT & DETAILS OF ROOF & GRADE SLAB	MD-216-1800-EG-CV-GAD-1566
5.) GEOTECHNICAL REPORT	MD-216-3000-EG-CV-RPT-1008
6.) STRUCTURAL DESIGN CRITERIA	MD-216-0000-EG-ST-CRT-0001

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ATTACHMENT 1 STAAD INPUT DATA

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 18-Oct-19
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 2 5.65 0 0; 4 0 3.29961 0; 5 5.65 3.29961 0; 7 0 0 5.765;
10 0 3.29961 5.765; 23 0 -1.5 0; 24 5.65 -1.5 0; 26 0 -1.5 5.765;
28 11.25 3.29961 0; 29 11.25 0 0; 32 11.25 -1.5 0; 34 0 0 9.47; 35 5.65 0 9.47;
36 0 3.29961 9.47; 37 5.65 3.29961 9.47; 38 0 -1.5 9.47; 39 5.65 -1.5 9.47;
40 11.25 3.29961 9.47; 41 11.25 0 9.47; 42 11.25 -1.5 9.47; 43 18.56 3.29961 0;
44 18.56 0 0; 47 18.56 -1.5 0; 49 18.56 3.29961 9.47; 50 18.56 0 9.47;
51 18.56 -1.5 9.47; 52 25.76 3.29961 0; 53 25.76 0 0; 56 25.76 -1.5 0;
58 25.76 3.29961 9.47; 59 25.76 0 9.47; 60 25.76 -1.5 9.47; 61 31.36 3.29961 0;
62 31.36 0 0; 65 31.36 -1.5 0; 67 31.36 3.29961 9.47; 68 31.36 0 9.47;
69 31.36 -1.5 9.47; 70 36.54 3.29961 0; 71 36.54 0 0; 72 36.54 3.29961 5.765;
73 36.54 0 5.765; 74 36.54 -1.5 0; 75 36.54 -1.5 5.765; 76 36.54 3.29961 9.47;
77 36.54 0 9.47; 78 36.54 -1.5 9.47;
MEMBER INCIDENCES
1 4 5; 3 1 4; 4 2 5; 8 7 10; 11 4 10; 25 1 2; 30 7 1; 31 1 23; 32 2 24;
34 7 26; 478 5 28; 479 29 28; 483 2 29; 485 29 32; 488 7 34; 490 10 36;
497 36 37; 498 34 36; 499 35 37; 500 35 34; 501 34 38; 502 35 39; 503 37 40;
504 41 40; 505 41 35; 506 41 42; 507 28 43; 508 29 44; 513 40 49; 514 41 50;
516 44 43; 519 44 47; 524 50 49; 525 50 51; 526 43 52; 527 44 53; 532 49 58;
533 50 59; 535 53 52; 538 53 56; 543 59 58; 544 59 60; 545 52 61; 546 53 62;
551 58 67; 552 59 68; 554 62 61; 557 62 65; 562 68 67; 563 68 69; 564 61 70;
565 62 71; 570 67 76; 571 68 77; 573 71 70; 574 73 72; 575 70 72; 576 71 74;
577 73 75; 578 71 73; 579 72 76; 580 73 77; 581 77 76; 582 77 78;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17184e+007
POISSON 0.17
DENSITY 23.6158
ALPHA 5e-006
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27578.9
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
3 4 8 498 499 554 562 573 574 581 PRIS YD 0.25 ZD 0.6
1 478 497 503 545 551 564 570 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
```


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```
25 30 483 488 500 505 546 552 565 571 578 580 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
11 490 575 579 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
507 513 526 532 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
508 514 527 533 PRIS YD 0.6 ZD 0.25
485 506 519 525 538 544 PRIS YD 0.35 ZD 0.6
31 32 34 479 501 502 504 516 524 535 543 557 563 576 577 582 PRIS YD 0.3 ZD 0.6
CONSTANTS
BETA 90 MEMB 4 32 479 485 499 502 504 506 516 519 524 525 535 538 543 544 -
554 557 562 563
MATERIAL CONCRETE ALL
SUPPORTS
23 24 26 32 38 39 42 47 51 56 60 65 69 74 75 78 FIXED
**************************************************************************
MEMBER CRACKED
3 4 8 31 32 34 479 485 498 499 501 502 504 506 516 519 524 525 535 538 543 -
544 554 557 562 563 573 574 576 577 581 -
582 REDUCTION RIY 0.700000 RIZ 0.700000
1 11 25 30 478 483 488 490 497 500 503 505 507 508 513 514 526 527 532 533 -
545 546 551 552 564 565 570 571 575 578 TO 579 -
580 REDUCTION RIY 0.350000 RIZ 0.350000
MEMBER OFFSET
1 497 START 0.125 0 0
564 570 END -0.125 0 0
478 503 507 513 526 532 545 551 564 570 START 0.3 0 0
478 503 507 513 526 532 545 551 END -0.3 0 0
11 490 575 579 START 0 0 0.3
11 490 575 579 END 0 0 -0.3
DEFINE IBC 2003
SDS 0.1 SD1 0.03 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2
SELFWEIGHT 1
MEMBER WEIGHT
25 30 483 488 500 505 508 514 527 533 546 552 565 571 578 580 UNI 12
1 478 497 503 507 513 526 532 545 551 564 570 UNI 35
DEFINE WIND LOAD
TYPE 1 WIND X WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.500 M 40.000 M 10.000 M 2.000 0.010 0 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.893 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.64157 1.64157 HEIG  0  4.572
```


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TYPE 2 WIND X LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.500 M 40.000 M 11.000 M 2.000 0.010 1 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.892 -0.218 0.550
!> END GENERATED DATA BLOCK
INT -0.966664 -0.966664 HEIG 0 3.5
TYPE 3 SIDEWARD X
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.500 M 40.000 M 11.000 M 2.000 0.010 2 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.892 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.5244 -1.5244 HEIG 0 3.5
TYPE 4 WIND Z WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.500 M 11.000 M 40.000 M 2.000 0.010 0 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.863 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.61078 1.61078 HEIG 0 4.572
TYPE 5 WIND Z LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.500 M 11.000 M 40.000 M 2.000 0.010 1 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.863 -0.500 0.550
!> END GENERATED DATA BLOCK
INT -1.27453 -1.27453 HEIG 0 3.5
TYPE 6 SIDEWARD Z
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 3.500 M 11.000 M 40.000 M 2.000 0.010 2 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.863 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.4987 -1.4987 HEIG 0 3.5
LOAD 1 LOADTYPE Seismic TITLE EQK LOAD Z
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
LOAD 2 LOADTYPE Seismic TITLE EQK LOAD X

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IBC LOAD X 1
PERFORM ANALYSIS
CHANGE
LOAD 3 LOADTYPE Dead TITLE DL
SELFWEIGHT Y -1
MEMBER LOAD
25 30 483 488 500 505 508 514 527 533 546 552 565 571 578 580 UNI GY -12
1 478 497 503 507 513 526 532 545 551 564 570 UNI GY -35
11 490 UMOM GZ 1
575 579 UMOM GZ -1
1 478 507 526 545 564 UMOM GX -1
497 503 513 532 551 570 UMOM GX 1
LOAD 4 LOADTYPE Live TITLE LL
MEMBER LOAD
1 478 497 503 507 513 526 532 545 551 564 570 UNI GY -2.42
11 490 UMOM GZ 0.1
575 579 UMOM GZ -0.1
1 478 507 526 545 564 UMOM GX -0.1
497 503 513 532 551 570 UMOM GX 0.1
LOAD 5 LOADTYPE Wind TITLE WL+X
WIND LOAD X 1 TYPE 1 YR 0 3.5
WIND LOAD -X 1 TYPE 2 YR 0 3.5
WIND LOAD -Z 1 TYPE 3 YR 0 3.5
WIND LOAD -Z -1 TYPE 3 YR 0 3.5
LOAD 6 LOADTYPE Wind TITLE WL+Z
WIND LOAD Z 1 TYPE 4 YR 0 3.5
WIND LOAD -Z 1 TYPE 5 YR 0 4
WIND LOAD -X 1 TYPE 6 YR 0 3.5
WIND LOAD -X -1 TYPE 6 YR 0 3.5
LOAD 7 LOADTYPE Temperature TITLE TEMP
TEMPERATURE LOAD
1 3 4 8 11 478 479 490 497 TO 499 503 504 507 513 516 524 526 532 535 543 -
545 551 554 562 564 570 573 TO 575 579 581 TEMP 25
LOAD COMB 8 SLS: (DL)
3 1.0
LOAD COMB 9 SLS: (DL+T)
3 1.0 7 1.0
LOAD COMB 10 SLS: (D+LR)
3 1.0 4 1.0
LOAD COMB 11 SLS: (D+0.75LR)
3 1.0 4 0.75
LOAD COMB 12 SLS: (D+0.7EZ)
3 1.0 1 0.7
LOAD COMB 13 SLS: (D+0.7EX)
3 1.0 2 0.7

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LOAD COMB 14 SLS: (D+0.75WX+0.75LR)
3 1.0 5 0.75 4 0.75
LOAD COMB 15 SLS: (D+0.75WZ+0.75LR)
3 1.0 6 0.75 4 0.75
LOAD COMB 16 SLS: (0.6D+WX)
3 0.6 5 1.0
LOAD COMB 17 SLS: (0.6D+WZ)
3 0.6 6 1.0
LOAD COMB 18 SLS: (0.6D+0.7EX)
3 0.6 2 0.7
LOAD COMB 19 SLS: (0.6D+0.7EZ)
3 0.6 1 0.7
LOAD COMB 20 SLS: (D+WX)
3 1.0 5 1.0
LOAD COMB 21 SLS: (D+WZ)
3 1.0 6 1.0
LOAD COMB 22 ULS: (1.4 DL)
3 1.4
LOAD COMB 23 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5 7 1.4
LOAD COMB 24 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5 7 1.4
LOAD COMB 25 ULS: (1.2D+0.80WX+1.6LR)
3 1.2 5 0.8 4 1.6
LOAD COMB 26 ULS: (1.2D+1.6WZ+0.5LR)
3 1.2 6 1.6 4 0.5
LOAD COMB 27 ULS: (1.2D+1.6WX+0.5LR)
3 1.2 4 0.5 5 1.6
LOAD COMB 28 ULS: (1.2D+1.6WZ+0.5LR)
3 1.2 4 0.5 6 1.6
LOAD COMB 29 ULS: (1.2D+EX)
3 1.2 2 1.0
LOAD COMB 30 ULS: (1.2D+EZ)
3 1.2 1 1.0
LOAD COMB 31 ULS: (0.90D+1.6WX)
3 0.9 5 1.6
LOAD COMB 32 ULS: (0.90D+1.6WZ)
3 0.9 6 1.6
LOAD COMB 33 ULS: (0.90D+EZ)
3 0.9 1 1.0
LOAD COMB 34 ULS: (0.90D+EX)
3 0.9 2 1.0
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI

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```
FC 35000 ALL
CLB 0.04 ALL
CLS 0.04 ALL
CLT 0.04 ALL
MAXMAIN 20 MEMB 3 4 8 31 32 34 479 498 499 501 502 504 516 524 535 543 554 -
557 562 563 573 574 576 577 581 582
MINMAIN 20 MEMB 3 4 8 31 32 34 479 485 498 499 501 502 504 506 516 519 524 -
525 535 538 543 544 554 557 562 563 573 574 576 577 581 582
MAXMAIN 25 MEMB 1 11 31 32 34 478 485 490 497 501 TO 503 506 513 519 525 532 -
538 544 545 551 557 563 564 570 575 TO 577 579 582
MINMAIN 20 MEMB 1 11 478 490 497 503 507 513 526 532 545 551 564 570 575 579
MAXMAIN 16 MEMB 11 490 575 579
MINMAIN 16 MEMB 11 490 575 579
MAXMAIN 16 MEMB 25 30 483 488 500 505 508 514 527 533 546 552 565 571 578 580
MINMAIN 16 MEMB 25 30 483 488 500 505 508 514 527 533 546 552 565 571 578 580
MINSEC 10 ALL
SFACE 0.5 MEMB 1 478 497 503 507 513 526 532 545 551 564 570
EFACE 0.5 MEMB 1 478 497 503 507 513 526 532 545 551 564 570
DESIGN BEAM 1 11 25 30 478 483 488 490 497 500 503 505 507 508 513 514 526 -
527 532 533 545 546 551 552 564 565 570 571 575 578 TO 580
DESIGN COLUMN 3 4 8 31 32 34 479 485 498 499 501 502 504 506 516 519 524 525 -
535 538 543 544 554 557 562 563 573 574 576 577 581 582
END CONCRETE DESIGN
FINISH
```


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ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS

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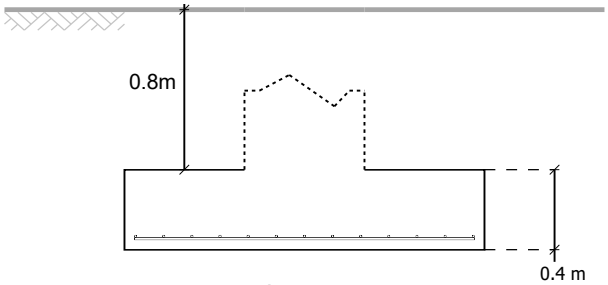
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Isolated Footing Design(ACI 318-08)

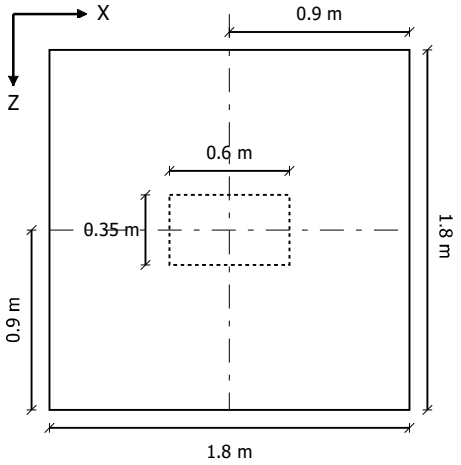
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
47	1	1.800m	1.800m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
47	13 - ϕ12	13 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F1 ALONG GRID 1 & D



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1800.000mm
Footing Width - Z (Fw) : 1800.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.600m

Column Width - Z (B_{col}) : 0.350m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.800m

Initial Width (W_o) = 1.800m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
9	SLS: (DL+T)	1.00	1.00	1.00
10	SLS: (D+LR)	1.00	1.00	1.00
11	SLS: (D+0.75LR)	1.00	1.00	1.00
12	SLS: (D+0.7EZ)	1.00	1.00	1.00
13	SLS: (D+0.7EX)	1.00	1.00	1.00
14	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
15	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
16	SLS: (0.6D+WX)	1.00	1.00	1.00
17	SLS: (0.6D+WZ)	1.00	1.00	1.00
18	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
19	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
22	ULS: (1.4 DL)	1.00	1.00	1.00
23	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
24	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
25	ULS: (1.2D+0.80WX+1.6LR)	1.00	1.00	1.00
26	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+1.6WX+0.5LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+EX)	1.00	1.00	1.00
30	ULS: (1.2D+EZ)	1.00	1.00	1.00
31	ULS: (0.90D+1.6WX)	1.00	1.00	1.00
32	ULS: (0.90D+1.6WZ)	1.00	1.00	1.00
33	ULS: (0.90D+EZ)	1.00	1.00	1.00
34	ULS: (0.90D+EX)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
9	383.863	-0.572	0.050	-5.728	0.342
10	403.557	0.393	0.049	-6.156	-2.238
11	399.449	0.413	0.048	-6.016	-2.222
12	387.133	0.391	10.101	35.483	-2.031
13	387.043	10.787	0.044	-5.596	-20.106
14	399.440	4.014	-26.883	-85.125	-7.919
15	399.380	0.228	28.494	77.757	-1.961
16	232.262	5.085	-35.880	-108.836	-8.901
17	232.184	0.037	37.955	108.340	-0.956
18	232.193	10.598	0.027	-3.358	-19.236
19	232.284	0.201	10.083	37.721	-1.160
20	387.112	5.275	-35.863	-111.073	-9.771
21	387.033	0.227	37.973	106.102	-1.827

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
22	541.974	0.664	0.062	-7.832	-3.046
23	545.624	-0.842	0.073	-8.300	0.448
24	545.624	-0.842	0.073	-8.300	0.448
25	490.832	4.279	-28.665	-91.995	-8.787
26	472.620	0.133	60.741	171.720	-2.083

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
27	472.746	8.209	-57.396	-175.761	-14.794
28	472.620	0.133	60.741	171.720	-2.083
29	464.433	15.302	0.053	-6.716	-28.226
30	464.562	0.450	14.420	51.969	-2.404
31	348.392	8.108	-57.411	-173.801	-14.111
32	348.266	0.031	60.726	173.679	-1.400
33	348.425	0.308	14.407	53.648	-1.751
34	348.296	15.160	0.040	-5.037	-27.573

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 3.240 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 2.474 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.800 m Governing Load Case : # 9

Width (W_2) = 1.800 m Governing Load Case : # 9

Depth (D_2) = 0.400 m Governing Load Case : # 29

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 3.240 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 31.103 kN

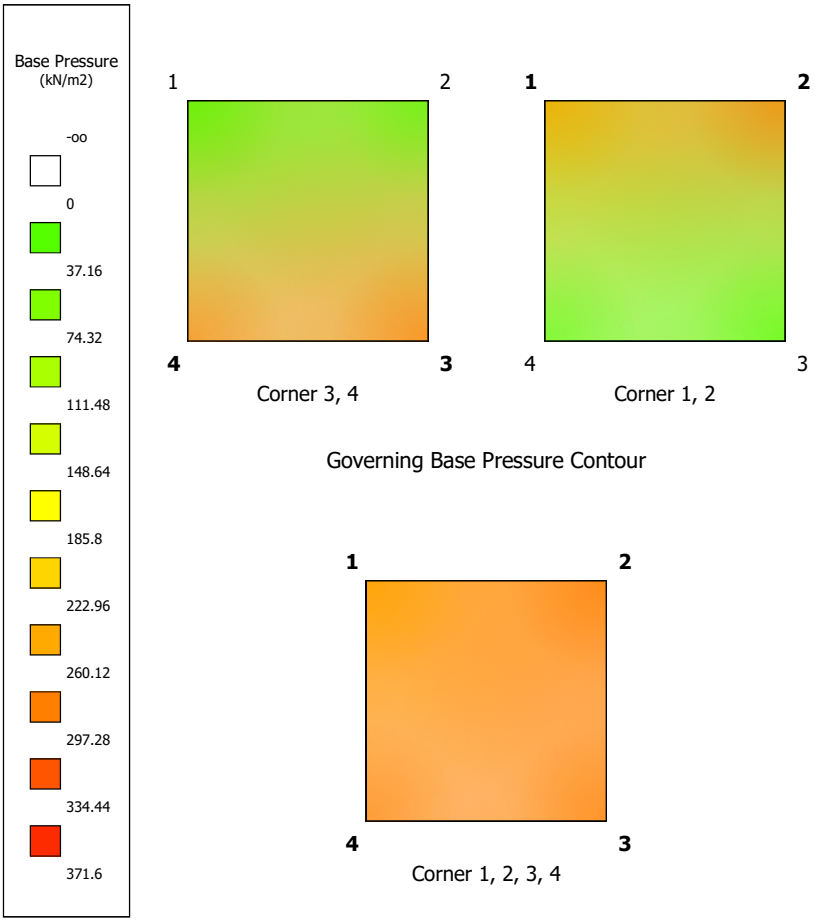
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 43.630 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



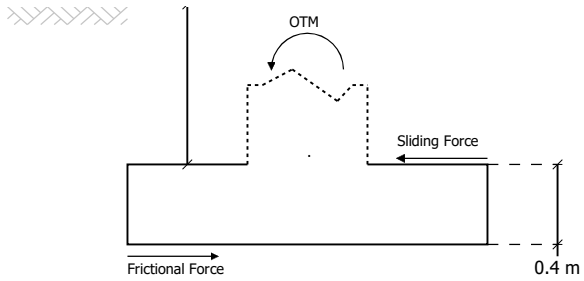
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
20	**259.3549**	283.8015	25.7349	1.2883	0.000
20	259.3549	**283.8015**	25.7349	1.2883	0.000
21	15.7609	19.7060	**269.2802**	265.3351	0.000
21	15.7609	19.7060	269.2802	**265.3351**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
20	**259.3549**	283.8015	25.7349	1.2883
20	259.3549	**283.8015**	25.7349	1.2883
21	15.7609	19.7060	**269.2802**	265.3351
21	15.7609	19.7060	269.2802	**265.3351**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
9	424.819	4829.248	598.474	1.500	72.643	723.684	1.500
10	645.806	5206.496	906.362	1.500	70.461	179.889	1.500
11	608.665	5282.979	855.126	1.500	71.489	178.902	1.500
12	626.457	24.234	34.246	1.500	10.566	190.255	1.500
13	22.688	5519.531	32.085	1.500	74.847	17.034	1.500
14	62.616	9.349	13.076	1.500	4.471	44.847	1.500
15	1104.500	8.819	12.471	1.500	4.808	208.166	1.500
16	31.996	4.535	6.350	1.500	2.258	25.303	1.500
17	4369.921	4.286	6.061	1.500	2.252	284.809	1.500
18	15.349	6103.524	21.707	1.500	83.088	11.784	1.500
19	808.972	16.137	22.817	1.500	6.664	223.006	1.500
20	46.404	6.825	9.550	1.500	3.330	35.018	1.500
21	1078.923	6.445	9.114	1.500	3.442	216.957	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 18
Governing Disturbing Force : 10.598kN
Governing Restoring Force : 162.671kN
Minimum Sliding Ratio for the Critical Load Case : 15.349
Critical Load Case for Overturning about X-Direction : 17
Governing Overturning Moment : 123.521kNm
Governing Resisting Moment : 278.148kNm
Minimum Overturning Ratio for the Critical Load Case : 2.252

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 17
Governing Disturbing Force : 37.955kN
Governing Restoring Force : 162.666kN
Minimum Sliding Ratio for the Critical Load Case : 4.286
Critical Load Case for Overturning about Z-Direction : 18
Governing Overturning Moment : -23.475kNm
Governing Resisting Moment : 276.618kNm
Minimum Overturning Ratio for the Critical Load Case : 11.784

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 17

Governing Disturbing Force : 37.955kN

Governing Restoring Force : 230.044kN

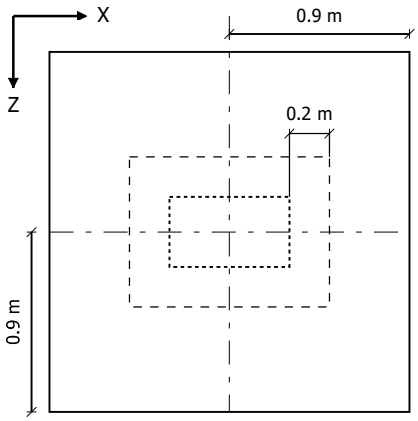
Minimum Sliding Ratio for the Critical Load Case : 6.061

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 1.714

Effective depth, d_{eff}, increased until 0.75XV_c $\geq$ Punching Shear Force

Punching Shear Force, Vu = 494.920kN, Load Case # 23

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.276m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2398.933kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3432.454kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_{.c}'} \times b_{.o} \times d_{.eff}$ 2214.399kN

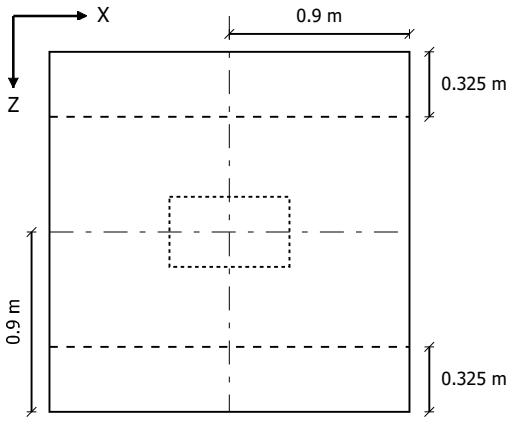
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1660.799kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 608.351kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.381m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

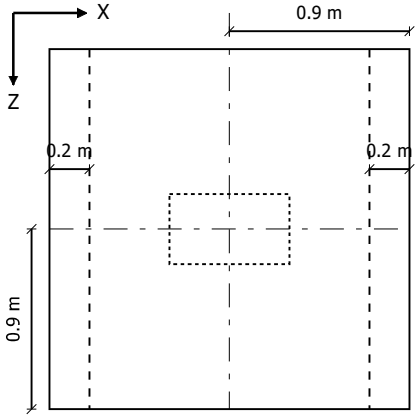
From above calculations, 0.75 X V_c = 456.264 kN

Critical load case for V_{ux} is # 27 $V_{ux} = V_{ux}|_{x=D_x}$ = 226.415 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 608.351 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.256 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

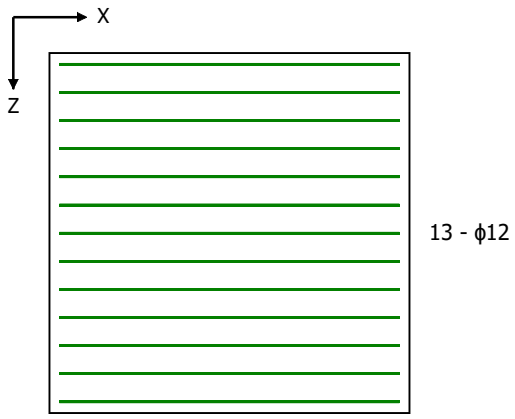
From above calculations, 0.75 X V_c = 456.264 kN

Critical load case for V_{uz} is # 29 $V_{uz} = V_{uz}|_{z=D_z}$ = 90.649 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 29

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00165
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.200 m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	62.837 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	69.818 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00078
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00067
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1190.735 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = =0.410 m

$$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}} =$$

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.550 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 13

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1470.255 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00237

From ACI Cl.7.6.1, minimum req'd clear distance between bars

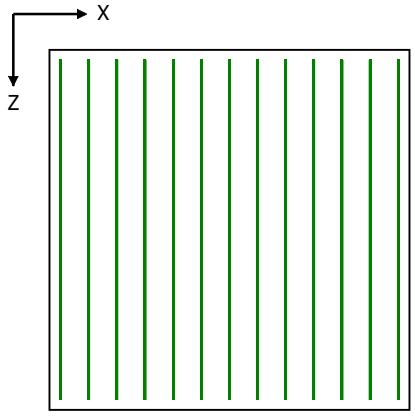
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.235

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



13 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 26

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00171

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 1.075 m

Ultimate moment, $M_u|_{x=D_x}$ = 154.130 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 171.255 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00209

(Based on gross depth) ρ x d_{eff} / Depth = 0.00173

Since $\rho_{min} < \rho < \rho_{max}$ OK

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1246.541 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.675 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 13

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1470.255 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

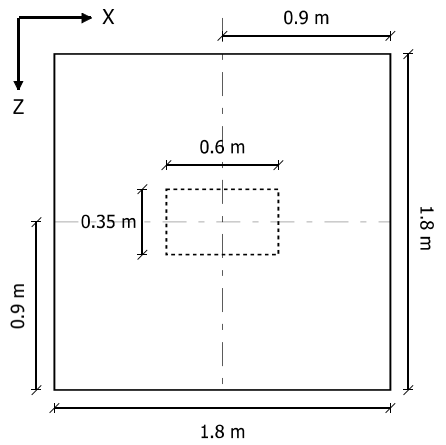
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00246

From ACI Cl.7.6.1, minimum req'd clear distance between bars

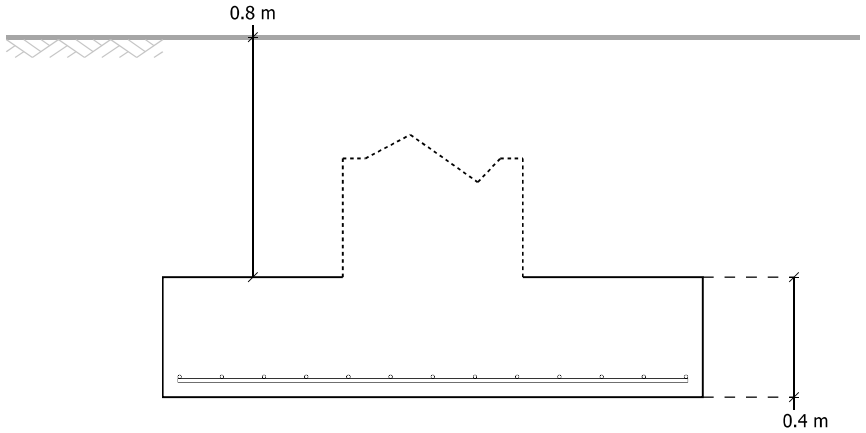
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.179

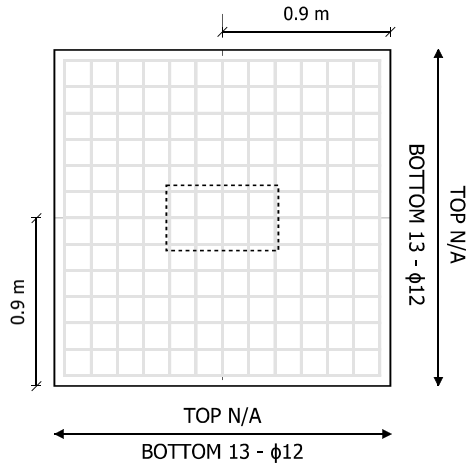
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.8 m	1.8 m	0.4 m	N/A	N/A	13 - ϕ12	13 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

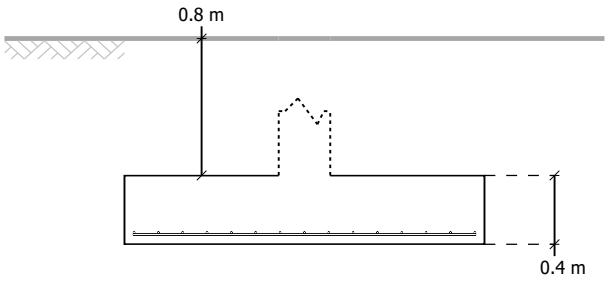
Print Calculation Sheet

Isolated Footing Design(ACI 318-08)

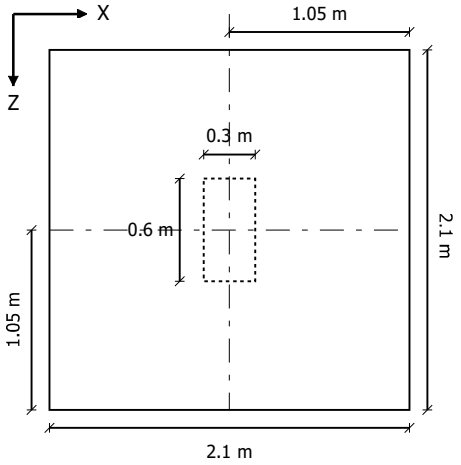
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
26	1	2.100m	2.100m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
26	15 - ϕ12	15 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F2 ALONG GRID 2 & A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 2100.000mm
Footing Width - Z (Fw) : 2100.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.300m

Column Width - Z (B_{col}) : 0.600m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 2.100m

Initial Width (W_o) = 2.100m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
9	SLS: (DL+T)	1.00	1.00	1.00
10	SLS: (D+LR)	1.00	1.00	1.00
11	SLS: (D+0.75LR)	1.00	1.00	1.00
12	SLS: (D+0.7EZ)	1.00	1.00	1.00
13	SLS: (D+0.7EX)	1.00	1.00	1.00
14	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
15	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
16	SLS: (0.6D+WX)	1.00	1.00	1.00
17	SLS: (0.6D+WZ)	1.00	1.00	1.00
18	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
19	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
22	ULS: (1.4 DL)	1.00	1.00	1.00
23	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
24	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
25	ULS: (1.2D+0.80WX+1.6LR)	1.00	1.00	1.00
26	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+1.6WX+0.5LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+EX)	1.00	1.00	1.00
30	ULS: (1.2D+EZ)	1.00	1.00	1.00
31	ULS: (0.90D+1.6WX)	1.00	1.00	1.00
32	ULS: (0.90D+1.6WZ)	1.00	1.00	1.00
33	ULS: (0.90D+EZ)	1.00	1.00	1.00
34	ULS: (0.90D+EX)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
9	102.131	-1.998	11.834	5.338	4.035
10	106.364	-1.099	12.321	6.194	-0.262
11	106.360	-1.091	12.320	6.190	-0.264
12	97.538	-0.458	18.680	16.122	-1.628
13	106.318	0.495	12.310	6.155	-3.574
14	104.786	13.619	12.247	5.904	-23.990
15	94.047	-11.424	22.938	21.683	14.621
16	61.710	18.972	7.292	3.325	-31.797
17	47.392	-14.418	21.547	24.364	19.684
18	63.778	0.922	7.384	3.684	-3.466
19	54.997	-0.030	13.754	13.651	-1.520
20	104.250	18.545	12.219	5.796	-31.905
21	89.932	-14.845	26.474	26.835	19.576

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
22	148.890	-1.495	17.243	8.648	-0.378
23	142.990	-2.812	16.570	7.481	5.653
24	142.990	-2.812	16.570	7.481	5.653
25	125.962	14.360	14.710	7.135	-25.619
26	101.358	-23.341	37.434	40.473	31.434

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
27	124.267	30.084	14.626	6.811	-50.936
28	101.358	-23.341	37.434	40.473	31.434
29	127.575	0.951	14.771	7.380	-5.045
30	115.031	-0.410	23.871	21.620	-2.265
31	92.355	30.420	10.929	4.950	-50.859
32	69.447	-23.005	33.736	38.611	31.511
33	83.126	-0.089	20.176	19.766	-2.184
34	95.670	1.272	11.076	5.527	-4.964

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 4.410 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.004 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 2.100 m Governing Load Case : # 9

Width (W_2) = 2.100 m Governing Load Case : # 9

Depth (D_2) = 0.400 m Governing Load Case : # 27

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 4.410 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 42.335 kN

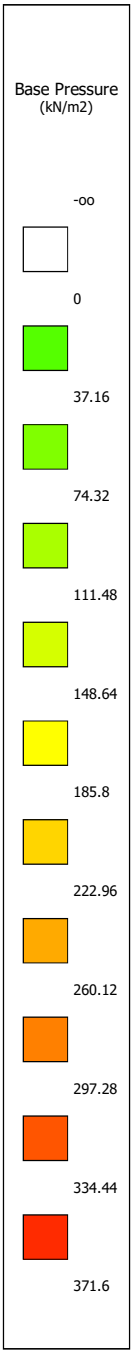
Gross Soil Bearing Capacity = 371.60 kN/m2

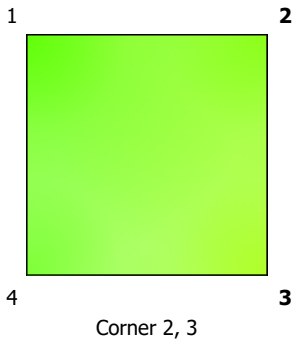
Soil Weight On Top Of Footing = 60.910 kN

Pressures at 4 Corners

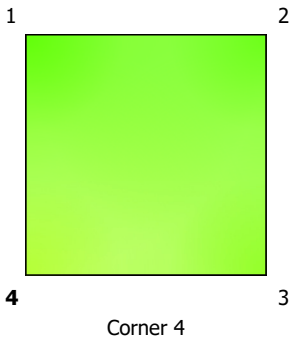
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

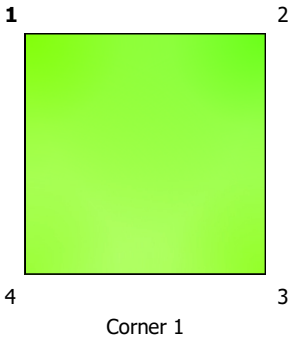




Corner 2, 3

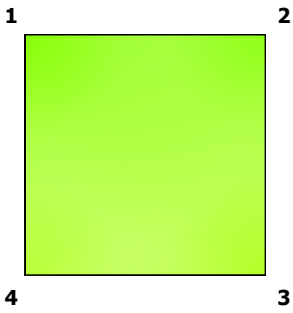


Corner 4



Corner 1

Governing Base Pressure Contour



Corner 1, 2, 3, 4

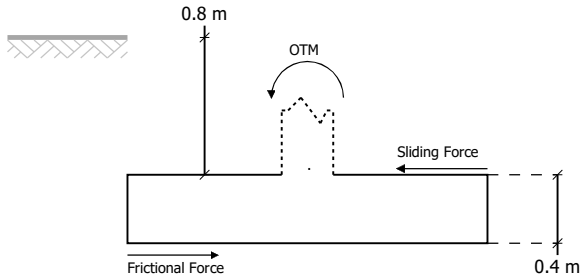
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
9	**43.1772**	36.9132	49.9636	56.2275	0.000
20	14.6522	**65.6059**	79.4497	28.4961	0.000
20	14.6522	65.6059	**79.4497**	28.4961	0.000
21	36.0880	3.0271	51.5205	**84.5814**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
9	**43.1772**	36.9132	49.9636	56.2275
20	14.6522	**65.6059**	79.4497	28.4961
20	14.6522	65.6059	**79.4497**	28.4961
21	36.0880	3.0271	51.5205	**84.5814**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
9	54.489	9.198	12.826	1.500	21.411	45.030	1.500
10	101.129	9.016	12.700	1.500	19.787	1251.006	1.500
11	101.839	9.017	12.702	1.500	19.795	1287.853	1.500
12	232.556	5.697	8.054	1.500	8.935	147.266	1.500
13	224.330	9.022	12.749	1.500	19.861	58.869	1.500
14	8.096	9.003	8.513	1.500	20.219	7.489	1.500
15	9.153	4.559	5.771	1.500	6.713	10.901	1.500
16	4.608	11.989	6.083	1.500	27.746	4.449	1.500
17	5.537	3.705	4.355	1.500	4.795	6.295	1.500
18	95.987	11.989	16.824	1.500	26.423	46.258	1.500
19	2752.754	6.098	8.624	1.500	8.675	111.511	1.500
20	5.930	9.000	7.003	1.500	20.392	5.592	1.500
21	6.897	3.867	4.770	1.500	5.420	8.030	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 16
Governing Disturbing Force : 18.972kN
Governing Restoring Force : 87.426kN
Minimum Sliding Ratio for the Critical Load Case : 4.608
Critical Load Case for Overturning about X-Direction : 17
Governing Overturning Moment : 32.982kNm
Governing Resisting Moment : 158.166kNm
Minimum Overturning Ratio for the Critical Load Case : 4.795

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 17

Governing Disturbing Force : 21.547kN

Governing Restoring Force : 79.837kN

Minimum Sliding Ratio for the Critical Load Case : 3.705

Critical Load Case for Overturning about Z-Direction : 16

Governing Overturning Moment : -39.386kNm

Governing Resisting Moment : 175.240kNm

Minimum Overturning Ratio for the Critical Load Case : 4.449

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 17

Governing Disturbing Force : 25.926kN

Governing Restoring Force : 112.907kN

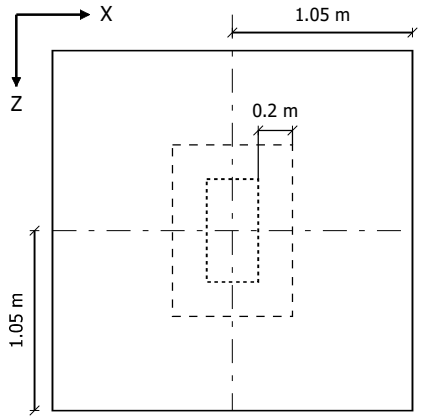
Minimum Sliding Ratio for the Critical Load Case : 4.355

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 217.377kN, Load Case # 22

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.176m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2146.805kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3398.657kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_{.c}'} \times b_{.o} \times d_{.eff}$ 2146.805kN

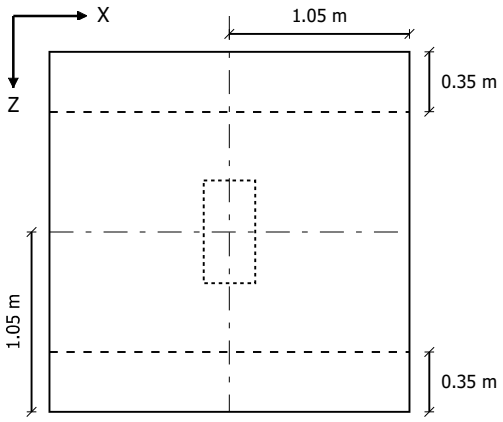
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1610.104kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 709.743kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.406m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

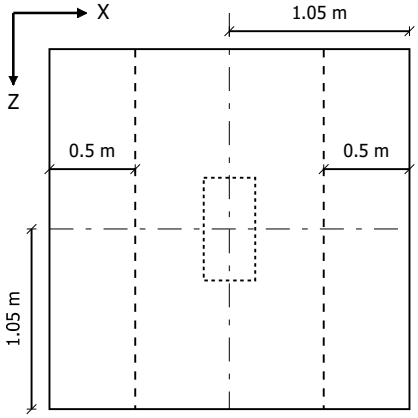
From above calculations, 0.75 X V_c = 532.308 kN

Critical load case for V_{ux} is # 26 $V_{ux} = V_{ux}|_{x=D_x}$ = 63.487 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 709.743 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.556 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

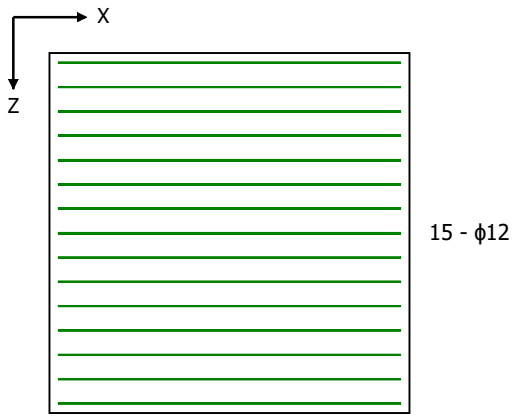
From above calculations, 0.75 X V_c = 532.308 kN

Critical load case for V_{uz} is # 27 $V_{uz} = V_{uz}|_{z=D_z}$ = 95.259 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 27

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl. 7.12.2, ρ_{min} =		0.00164	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.200	m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	68.679	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	76.310	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00073	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00063	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1380.565	mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = =0.410 m

$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.850 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 15

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1696.448 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00235

From ACI Cl.7.6.1, minimum req'd clear distance between bars

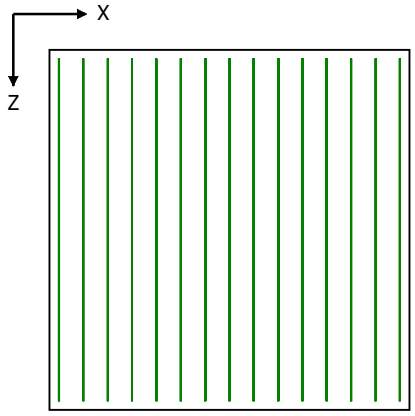
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.229

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



15 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 26

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00159

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 1.350 m

Ultimate moment, $M_u|_{x=D_x}$ = 44.550 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 49.500 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00051

(Based on gross depth) ρ x d_{eff} / Depth = 0.00042

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1335.422 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.700 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 15

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1696.448 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

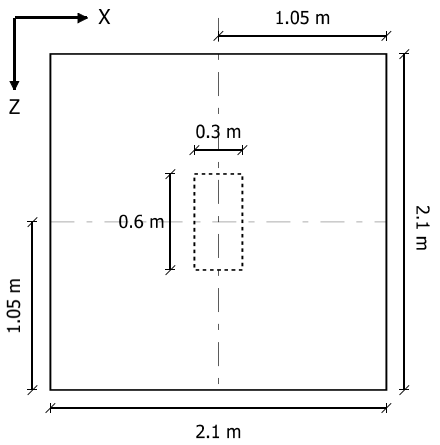
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00243

From ACI Cl.7.6.1, minimum req'd clear distance between bars

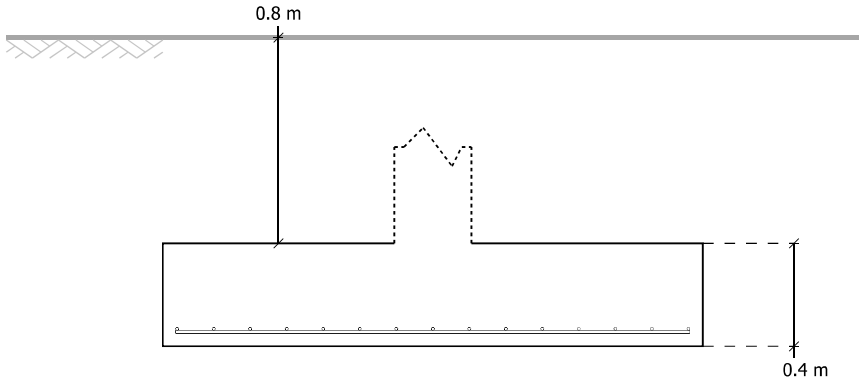
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.270

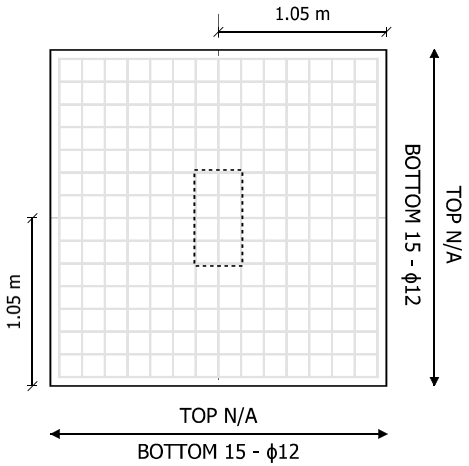
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

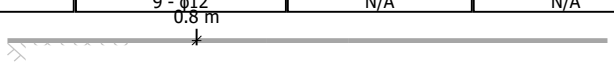
	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	2.1 m	2.1 m	0.4 m	N/A	N/A	15 - ϕ12	15 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

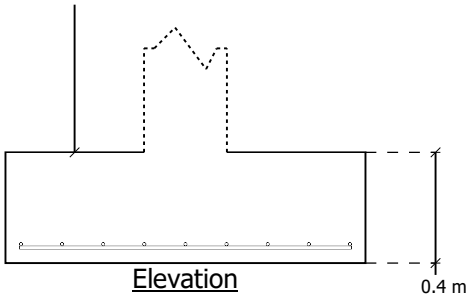
Isolated Footing Design(ACI 318-08)

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
38	1	1.300m	1.300m	0.400m

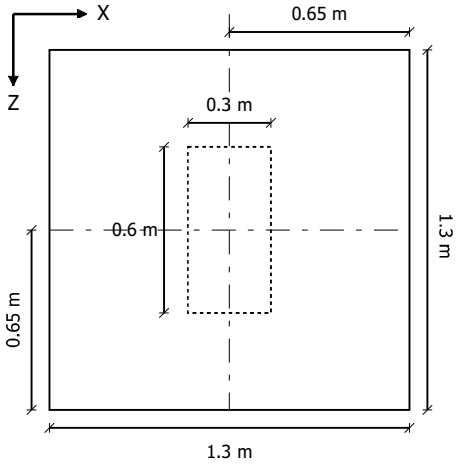
Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
38	9 - ϕ12	9 - ϕ12	N/A	N/A	N/A	N/A



FOUNDATION-F3 ALONG GRID 3 & A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1300.000mm
Footing Width - Z (Fw) : 1300.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.300m

Column Width - Z (B_{col}) : 0.600m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.300m

Initial Width (W_o) = 1.300m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
9	SLS: (DL+T)	1.00	1.00	1.00
10	SLS: (D+LR)	1.00	1.00	1.00
11	SLS: (D+0.75LR)	1.00	1.00	1.00
12	SLS: (D+0.7EZ)	1.00	1.00	1.00
13	SLS: (D+0.7EX)	1.00	1.00	1.00
14	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
15	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
16	SLS: (0.6D+WX)	1.00	1.00	1.00
17	SLS: (0.6D+WZ)	1.00	1.00	1.00
18	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
19	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
22	ULS: (1.4 DL)	1.00	1.00	1.00
23	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
24	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
25	ULS: (1.2D+0.80WX+1.6LR)	1.00	1.00	1.00
26	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+1.6WX+0.5LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+EX)	1.00	1.00	1.00
30	ULS: (1.2D+EZ)	1.00	1.00	1.00
31	ULS: (0.90D+1.6WX)	1.00	1.00	1.00
32	ULS: (0.90D+1.6WZ)	1.00	1.00	1.00
33	ULS: (0.90D+EZ)	1.00	1.00	1.00
34	ULS: (0.90D+EX)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
9	194.058	-10.547	1.204	1.630	9.908
10	187.476	-10.313	7.359	3.406	5.613
11	186.019	-10.395	7.355	3.405	5.620
12	192.337	-10.748	13.661	16.187	5.705
13	177.319	-6.463	7.334	3.371	0.515
14	184.795	-9.931	6.440	3.093	4.173
15	201.892	-10.370	18.505	23.583	6.064
16	107.357	-5.767	3.186	1.625	1.455
17	130.153	-6.352	19.272	28.946	3.977
18	104.659	-2.206	4.397	2.010	-1.742
19	119.677	-6.491	10.724	14.826	3.448
20	180.017	-10.024	6.123	2.987	3.712
21	202.813	-10.609	22.210	30.307	6.234

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
22	254.310	-14.900	10.280	4.764	7.899
23	274.594	-14.601	1.694	2.283	13.856
24	274.594	-14.601	1.694	2.283	13.856
25	225.995	-11.749	7.861	3.755	5.180
26	254.754	-12.553	32.606	47.131	7.703

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
27	218.281	-11.617	6.867	3.419	3.668
28	254.754	-12.553	32.606	47.131	7.703
29	211.794	-6.800	8.799	4.038	-0.554
30	233.248	-12.921	17.837	22.347	6.860
31	160.873	-8.589	4.657	2.396	1.990
32	197.346	-9.525	30.395	46.109	6.025
33	178.753	-9.728	15.634	21.326	5.167
34	157.299	-3.607	6.596	3.017	-2.246

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 1.690 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.265 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.300 m Governing Load Case : # 9

Width (W_2) = 1.300 m Governing Load Case : # 9

Depth (D_2) = 0.400 m Governing Load Case : # 26

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 1.690 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 16.223 kN

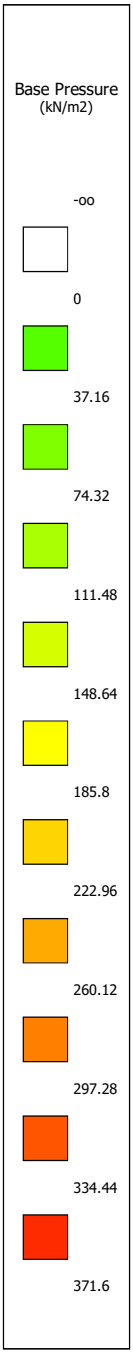
Gross Soil Bearing Capacity = 371.60 kN/m2

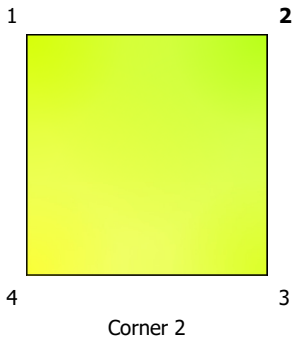
Soil Weight On Top Of Footing = 21.743 kN

Pressures at 4 Corners

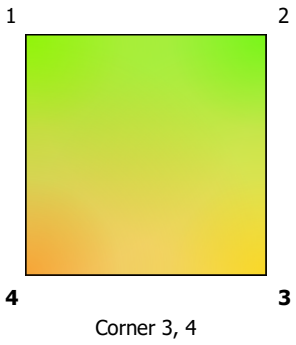
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

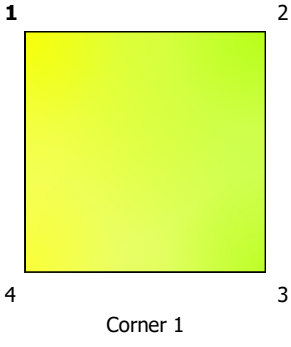




Corner 2

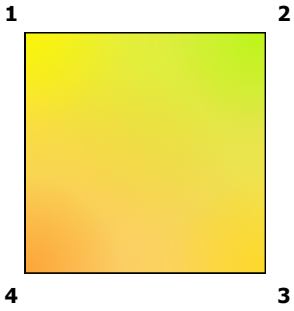


Corner 3, 4



Corner 1

Governing Base Pressure Contour



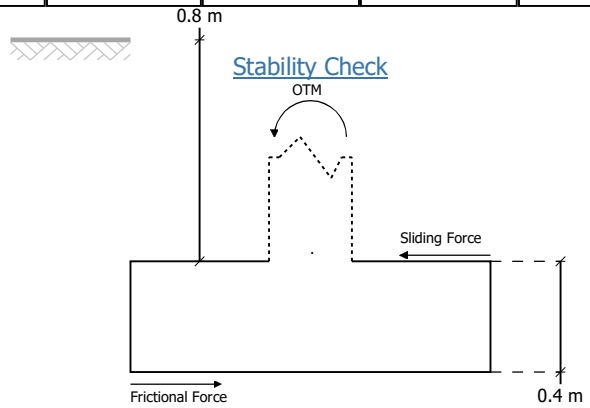
Corner 1, 2, 3, 4

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
9	**170.1065**	92.9461	104.4789	181.6393	0.000
13	118.6372	**101.7027**	136.1393	153.0737	0.000
21	64.0579	6.8271	**220.8883**	278.1191	0.000
21	64.0579	6.8271	220.8883	**278.1191**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
9	**170.1065**	92.9461	104.4789	181.6393
13	118.6372	**101.7027**	136.1393	153.0737
21	64.0579	6.8271	**220.8883**	278.1191
21	64.0579	6.8271	220.8883	**278.1191**



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
9	11.659	102.142	16.383	1.500	71.428	10.765	1.500
10	11.586	16.237	13.338	1.500	23.080	15.178	1.500
11	11.420	16.141	13.184	1.500	22.939	15.018	1.500
12	11.357	8.935	9.931	1.500	6.914	15.090	1.500
13	17.655	15.558	16.507	1.500	22.195	45.542	1.500
14	11.888	18.334	14.106	1.500	25.543	17.931	1.500
15	12.259	6.870	8.475	1.500	5.032	15.390	1.500
16	13.355	24.177	16.532	1.500	32.576	25.442	1.500
17	14.027	4.623	6.210	1.500	2.981	16.959	1.500
18	34.269	17.191	21.731	1.500	24.599	109.346	1.500
19	12.873	7.791	9.426	1.500	5.360	17.162	1.500
20	11.525	18.869	13.910	1.500	26.067	18.512	1.500
21	12.029	5.746	7.332	1.500	3.993	15.057	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 12
Governing Disturbing Force : -10.748kN
Governing Restoring Force : 122.061kN
Minimum Sliding Ratio for the Critical Load Case : 11.357
Critical Load Case for Overturning about X-Direction : 17
Governing Overturning Moment : 36.654kNm
Governing Resisting Moment : 109.276kNm
Minimum Overturning Ratio for the Critical Load Case : 2.981

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 17

Governing Disturbing Force : 19.272kN

Governing Restoring Force : 89.103kN

Minimum Sliding Ratio for the Critical Load Case : 4.623

Critical Load Case for Overturning about Z-Direction : 9

Governing Overturning Moment : 14.127kNm

Governing Resisting Moment : 152.077kNm

Minimum Overturning Ratio for the Critical Load Case : 10.765

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 17

Governing Disturbing Force : 20.292kN

Governing Restoring Force : 126.011kN

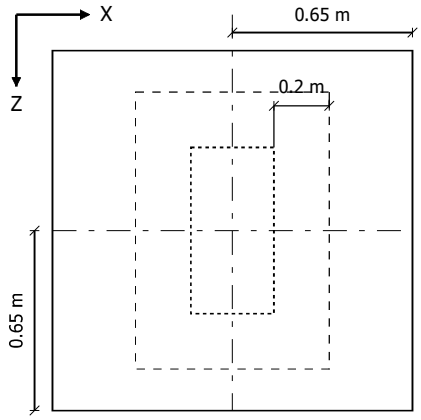
Minimum Sliding Ratio for the Critical Load Case : 6.210

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 200.125kN, Load Case # 23

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.176m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2146.805kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3398.657kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_{.c}'} \times b_{.o} \times d_{.eff}$ 2146.805kN

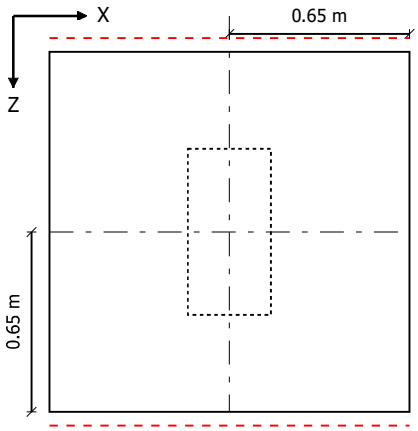
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1610.104kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 439.365kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.006m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

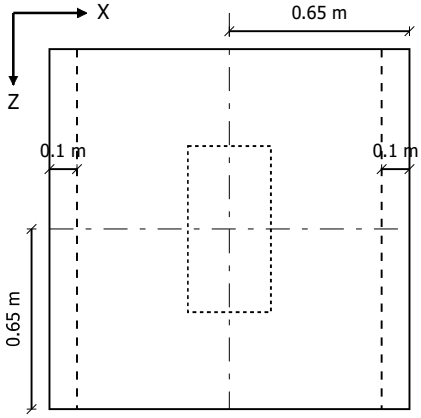
From above calculations, 0.75 X V_c = 329.524 kN

Critical load case for V_{ux} is # 26 $V_{ux} = V_{ux}|_{x=D_x}$ = 2.427 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 439.365 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.156 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

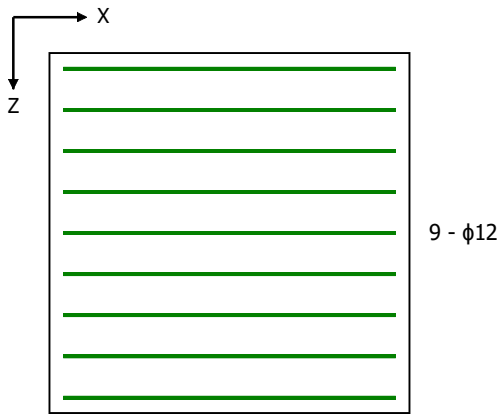
From above calculations, 0.75 X V_c = 329.524 kN

Critical load case for V_{uz} is # 23 $V_{uz} = V_{uz}|_{z=D_z}$ = 47.107 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 23

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl. 7.12.2, ρ_{min} =		0.00177	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.500	m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	36.551	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	40.613	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00063	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00054	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	918.298	mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 148.500mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 145mm o.c.

Required development length for bars = =0.410 m

$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =

Available development length for bars, D_L = $0.5 \times (L - D_{col}) \cdot - C_{cover}$ = 0.450 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 9

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1017.869 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00228

From ACI Cl.7.6.1, minimum req'd clear distance between bars

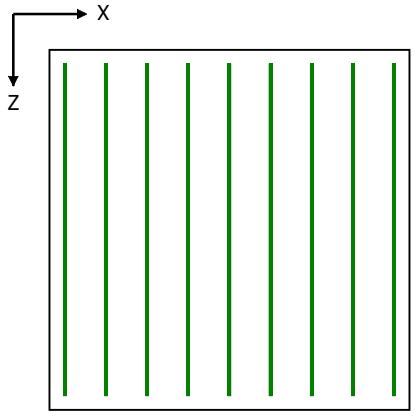
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.108

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



9 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 26

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00174

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.950 m

Ultimate moment, $M_u|_{x=D_x}$ = 24.754 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 27.505 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00046

(Based on gross depth) ρ x d_{eff} / Depth = 0.00038

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 902.299 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.300 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 9

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1017.869 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

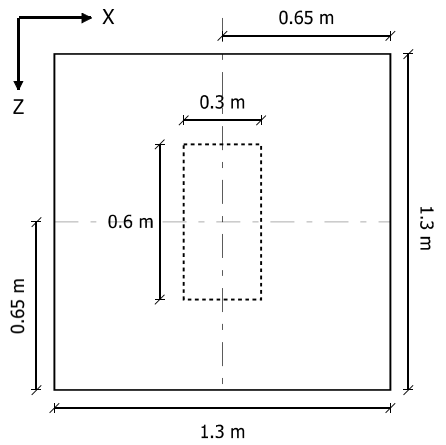
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00236

From ACI Cl.7.6.1, minimum req'd clear distance between bars

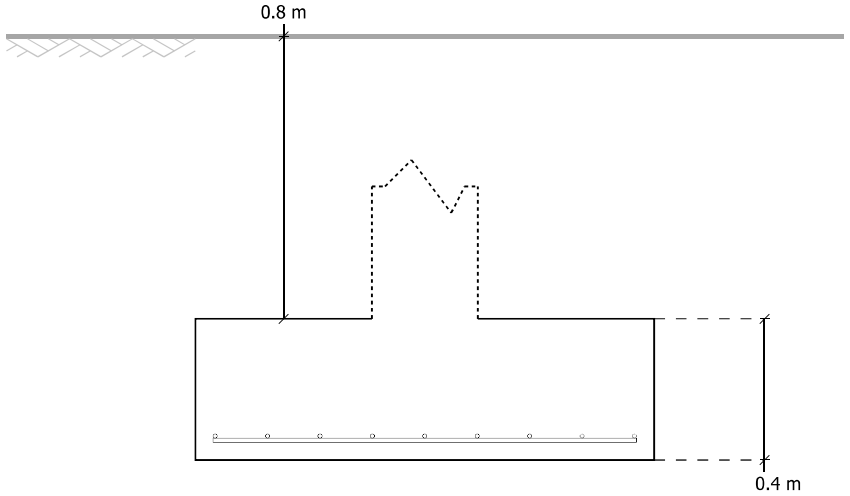
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.128

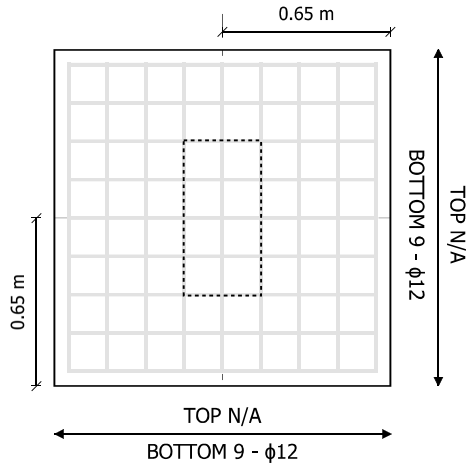
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.3 m	1.3 m	0.4 m	N/A	N/A	9 - ϕ12	9 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet